Computer Vision —— Object Recognition

2026/10/13

桑 农

Agenda

- Introduction
- Objects & Faces recognition
- Bag-of-words models
- Part-based models
- Deep Networks

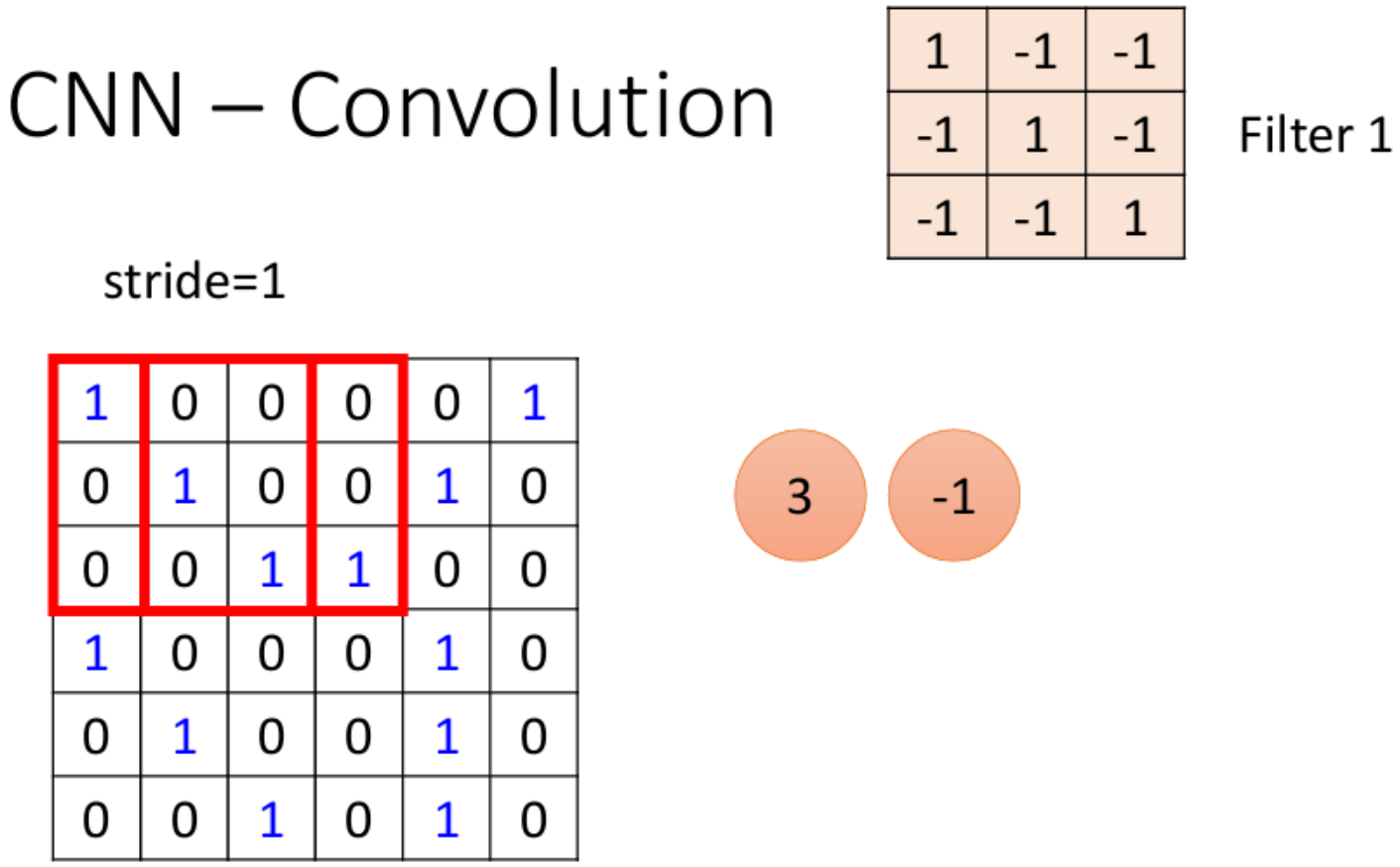
Deep Convolutional Networks

- Convolution
- Pooling
- Activation Function
- Softmax Activation Function
- Norm
- Loss Function
- Attention

Deep Convolutional Networks

- Convolution

  General

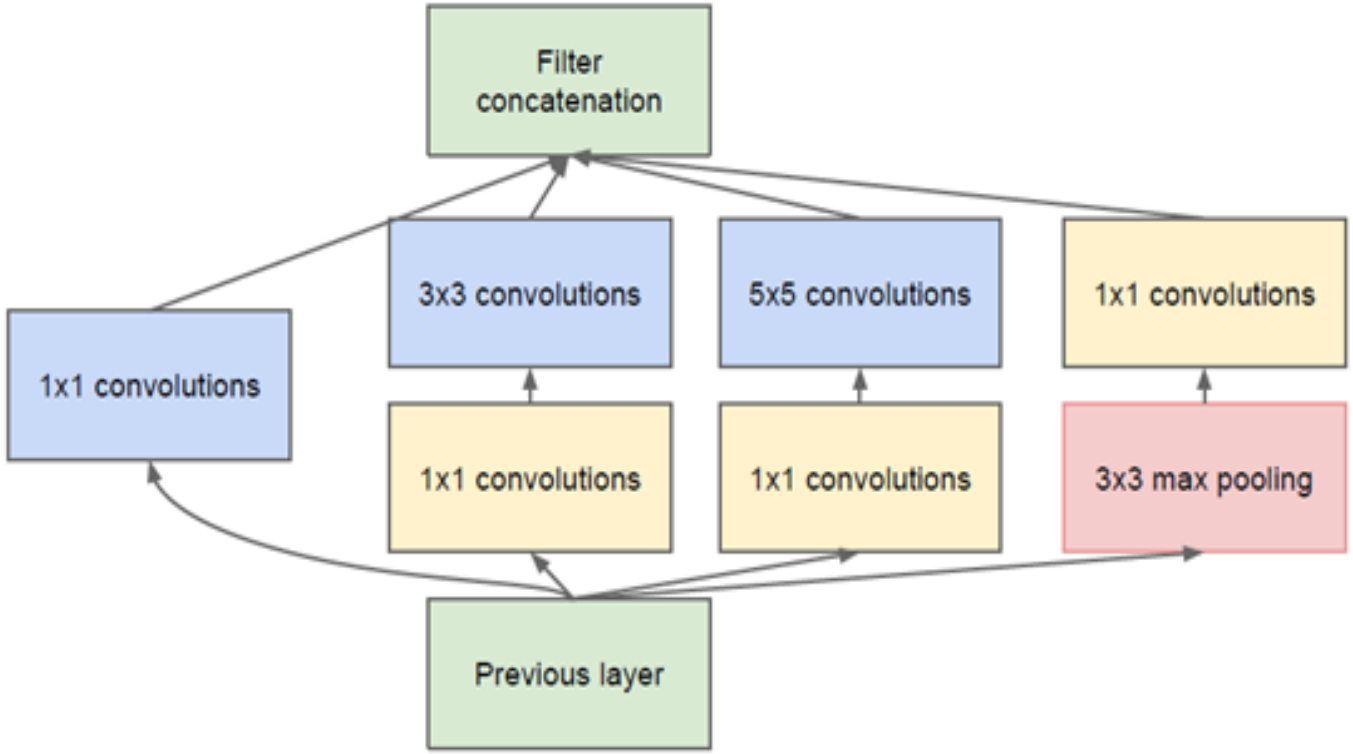


from Deep Learning Tutorial with 李宏毅

Deep Convolutional Networks

- Convolution

  Inception（GoogleNet）: 在一层中提取不同尺度的特征

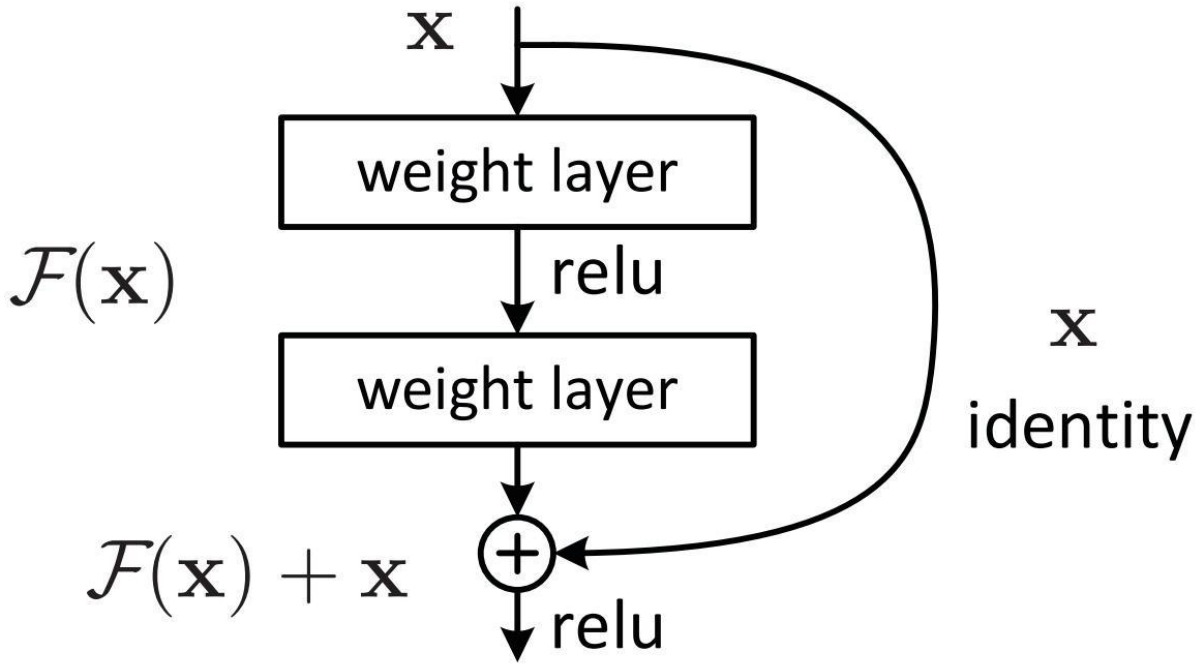


Inception V1

Deep Convolutional Networks

- Convolution

  ResNet:缓解反向传播算法的梯度消失和爆炸问题

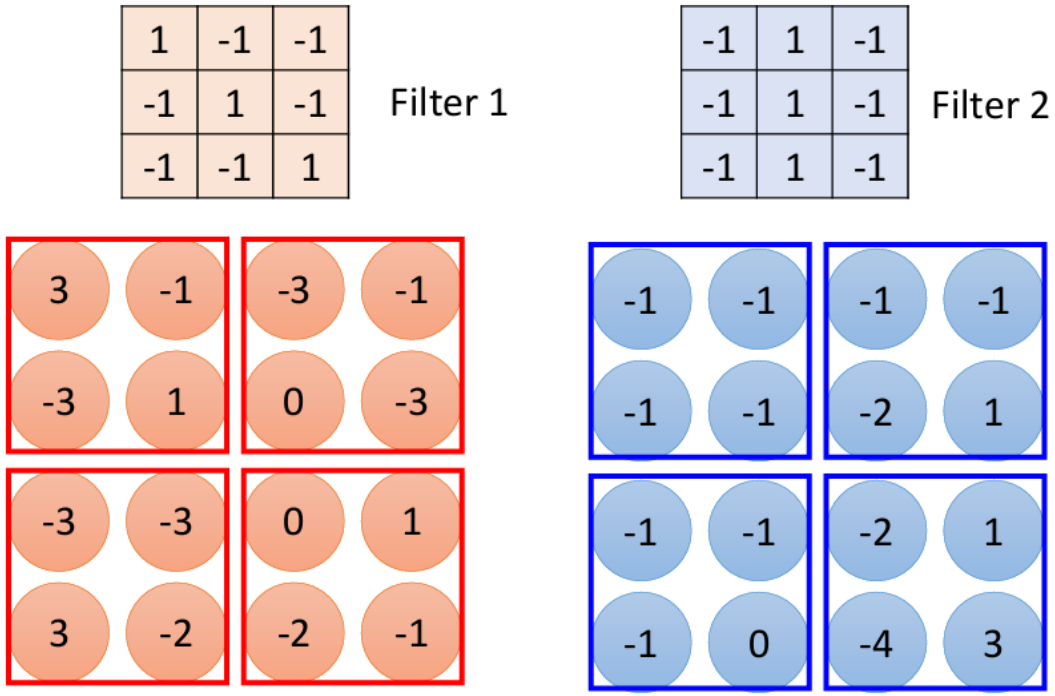


Deep Convolutional Networks

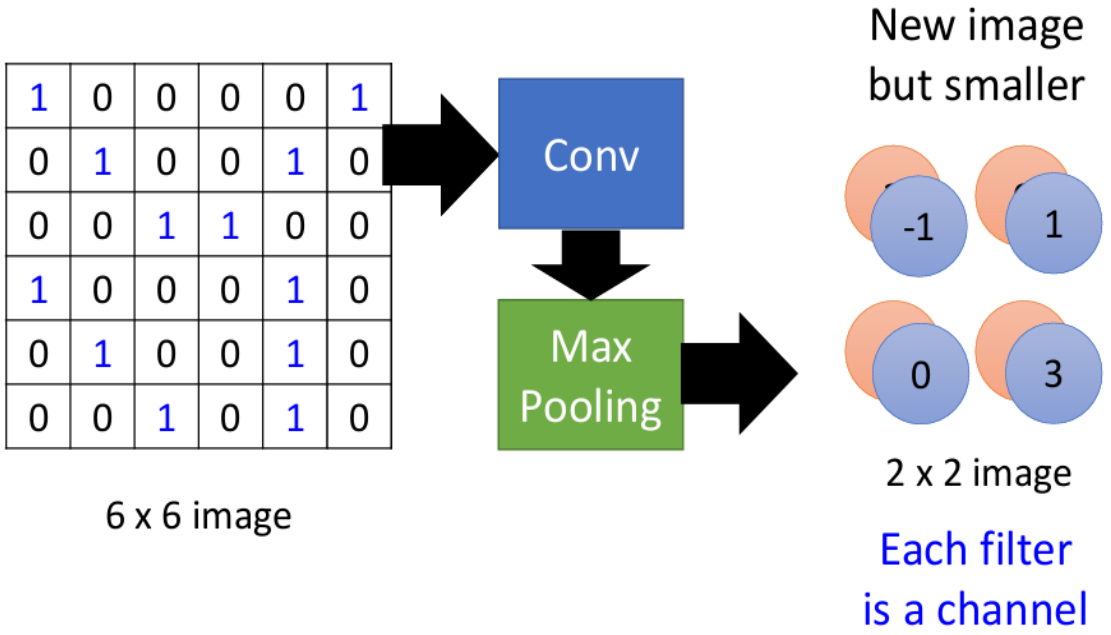
- Pooling

  Max Pooling

CNN – Max Pooling



CNN – Max Pooling



from Deep Learning Tutorial with 李宏毅

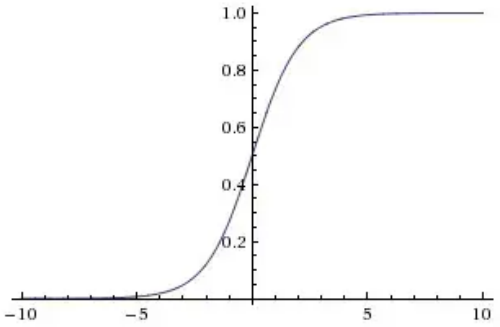
Average Pooling

Deep Convolutional Networks

- Activation Function

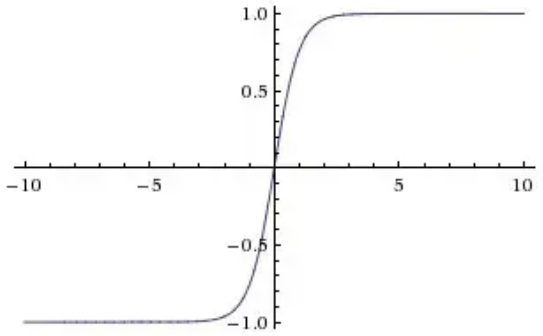
Sigmoid

$$f(x) = \frac{1}{1 + e^{-x}}$$



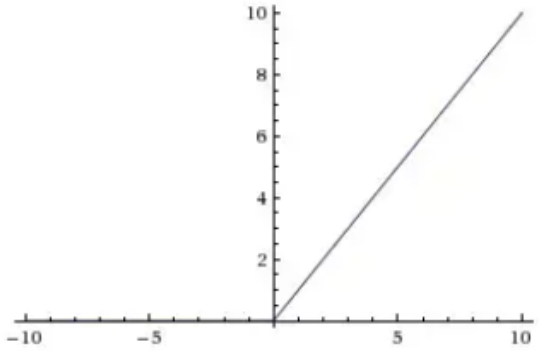
Tanh

$$f(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$



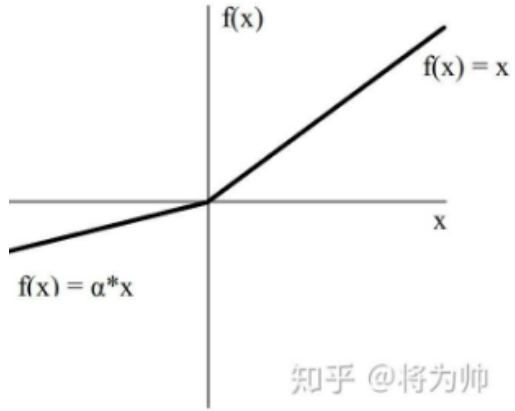
ReLU

$$f(x) = \max(0, x)$$



LeakyReLU

$$f(x) = \begin{cases} x & if\ x \geq 0 \\ \alpha x & if\ x < 0 \end{cases}$$

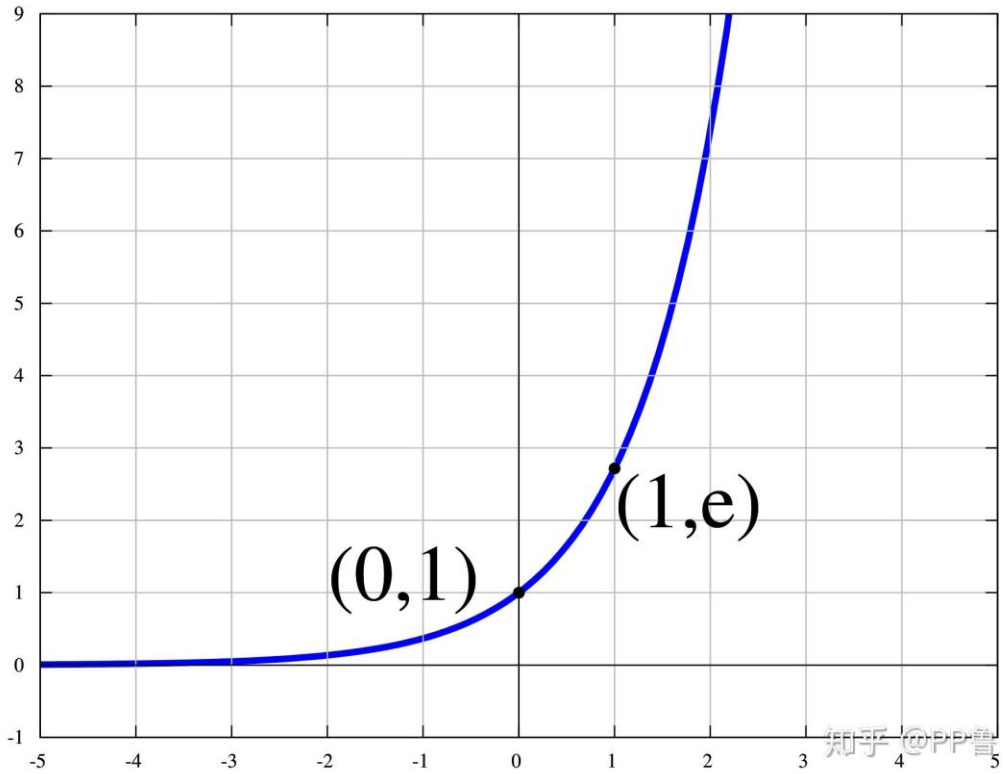


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Deep Convolutional Networks

- Softmax Activation Function

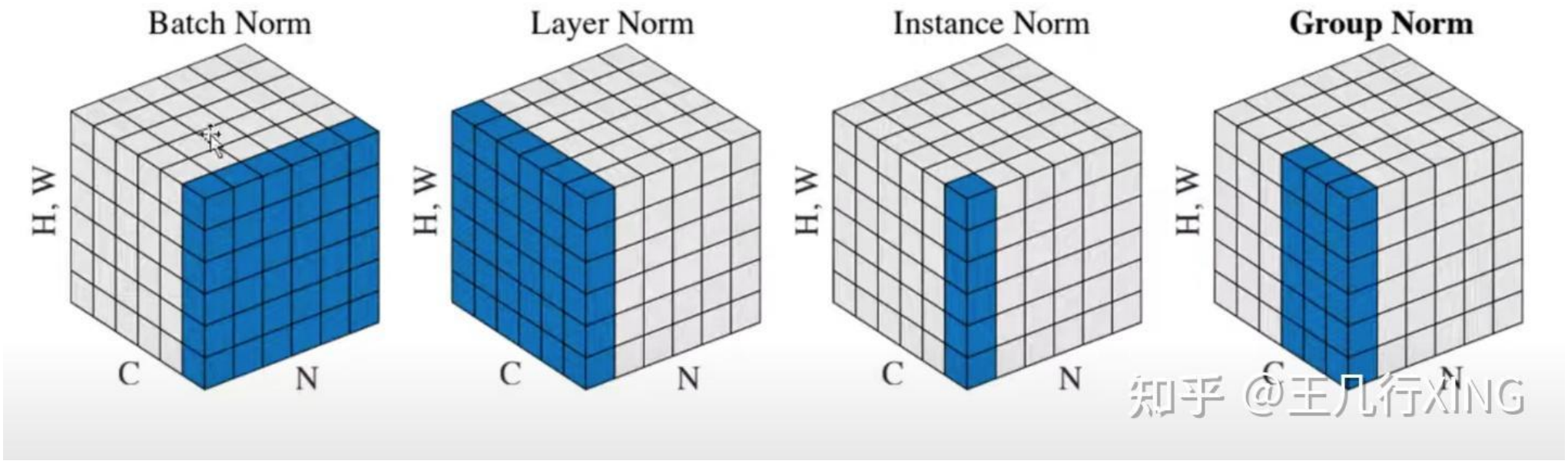
$$\text{Softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$



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Deep Convolutional Networks

- Norm



跨样本单通道　　单样本跨通道　　单样本单通道　　单样本跨若干通道

Deep Convolutional Networks

- Loss Fuction

Regression Loss

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Classification Loss

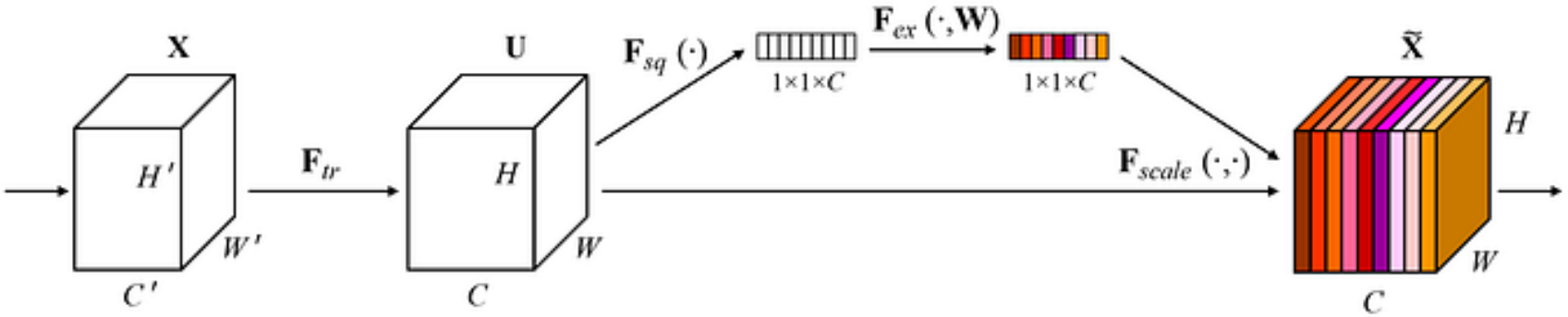
$$L = -\frac{1}{n}\sum_{i=1}^{n}\sum_{k}^{k-1} y_{ik}\log(\hat{y}_{ik})$$

$$L = -\frac{1}{n}\sum_{i=1}^{n}[y_i\log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)]$$

Deep Convolutional Networks

- Attention

  Channel Attention: Squeeze and Excitation Network (SENet)

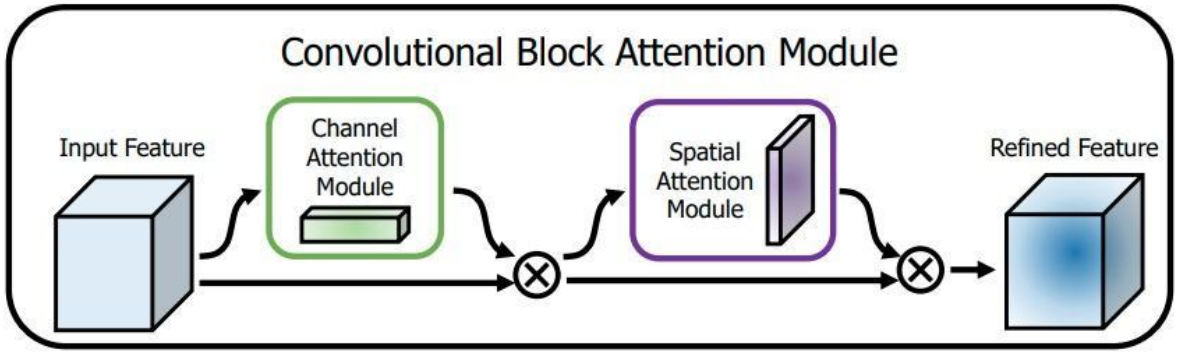


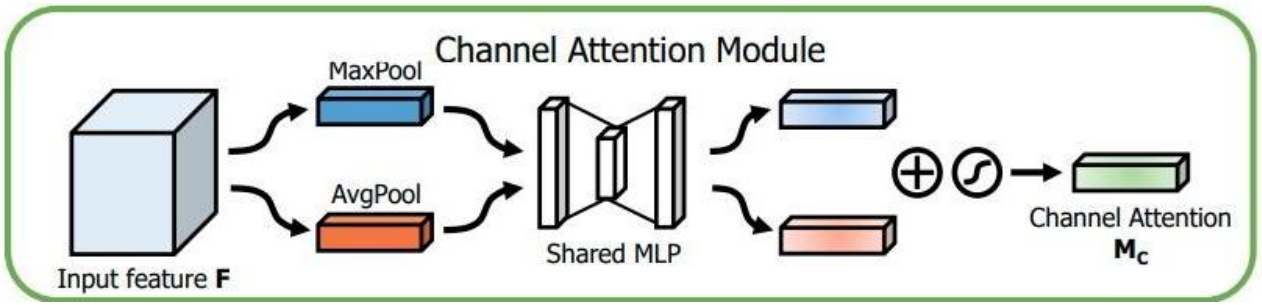
$$\sigma_i(y_i) = \frac{\exp(y_i)}{\sum_{j=1}^{n} \exp(y_j)}$$

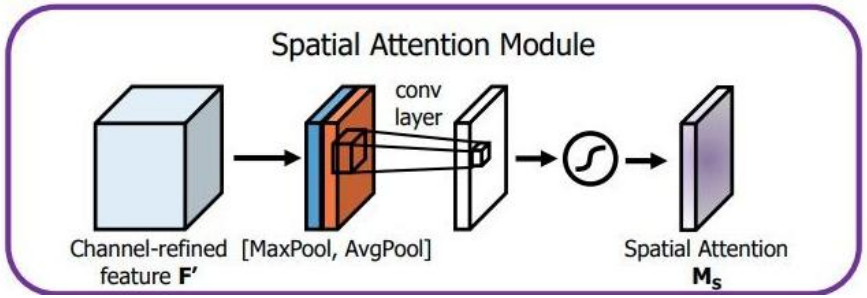
Deep Convolutional Networks

- Attention

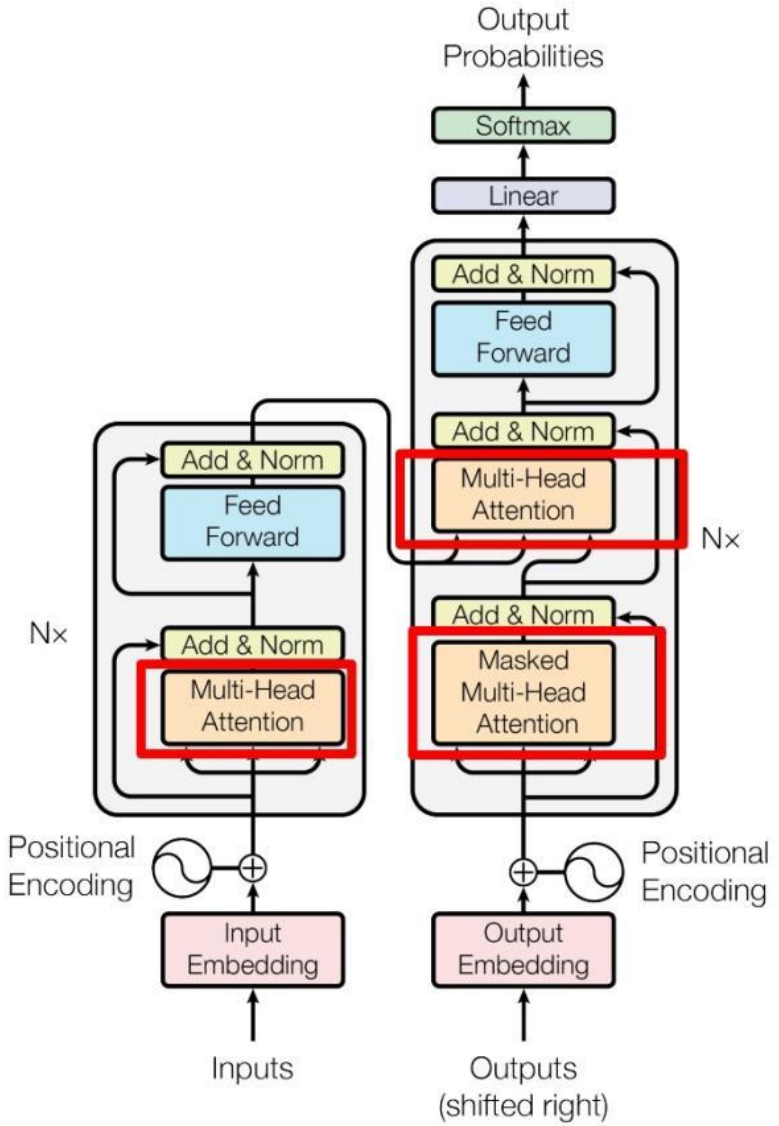
  Spatial-Channel Attention： Convolutional Block Attention Module（CBAM）



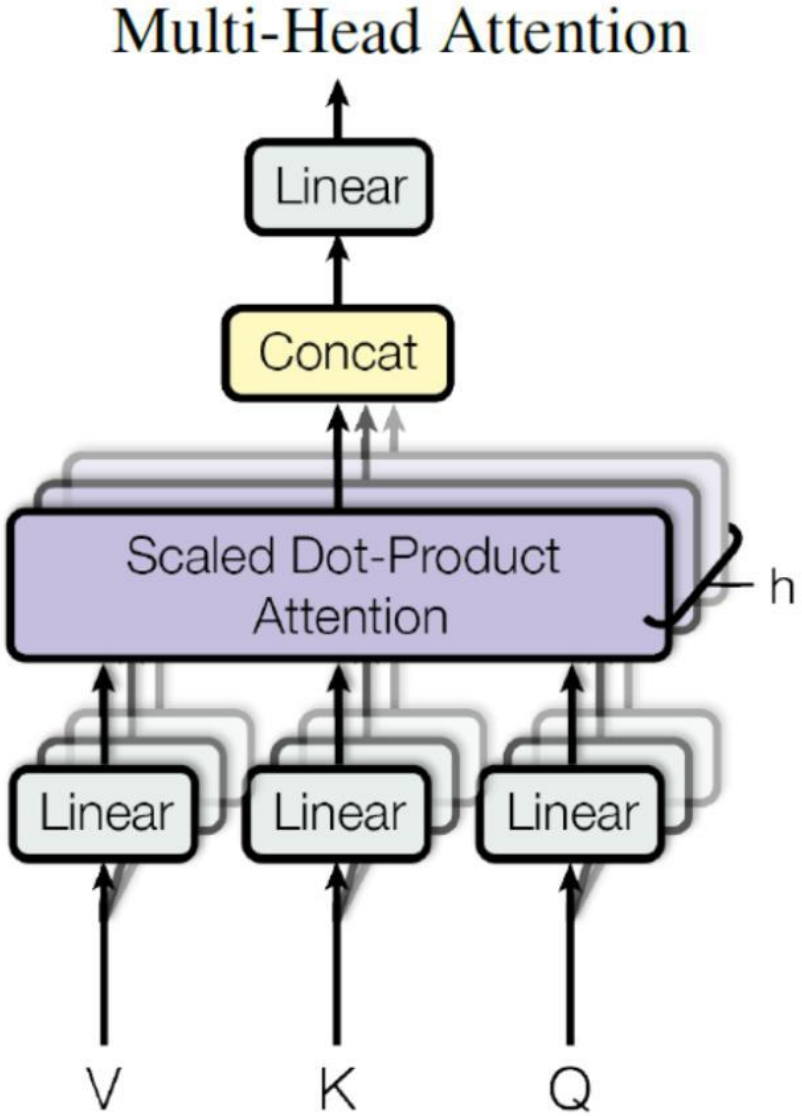


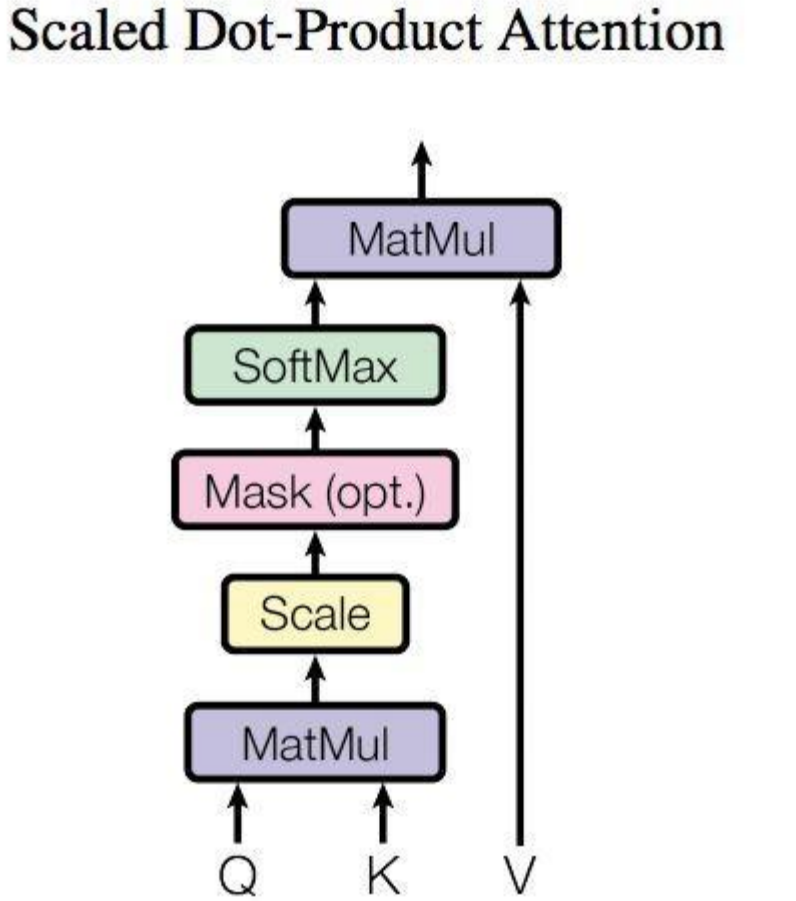


Transformer



Transformer





$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

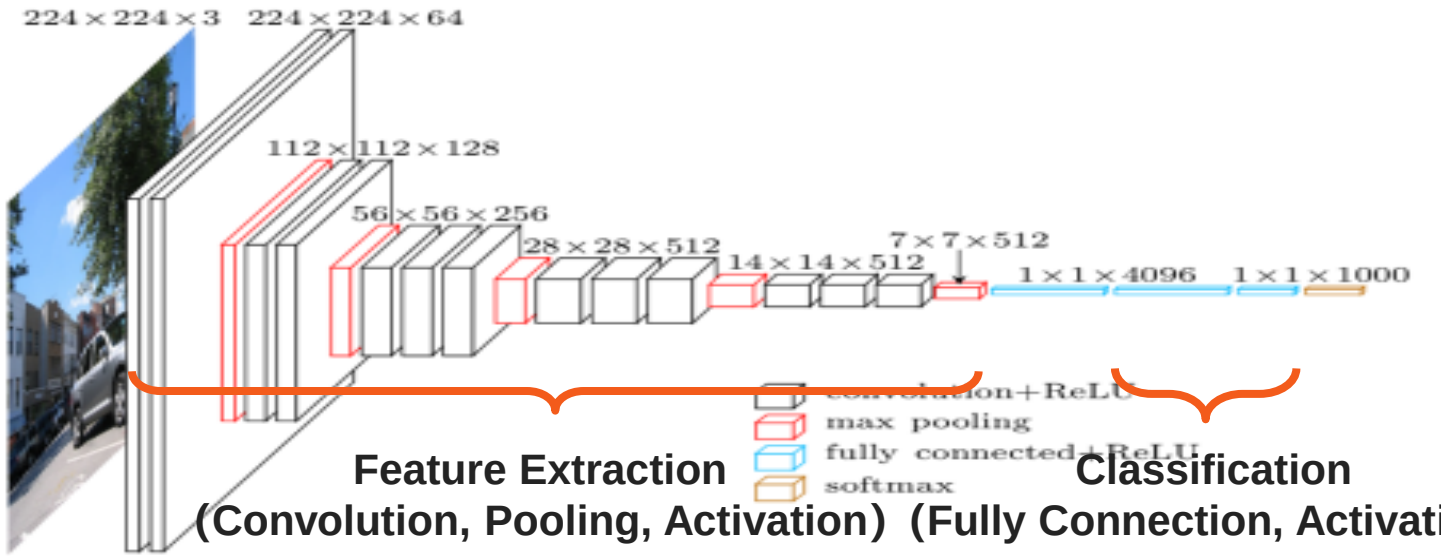
d_k是Q,K矩阵的列数，即向量维度

Recognition



Feature Extraction（HOG）

Classification（SVM, Linear Classifier, …）

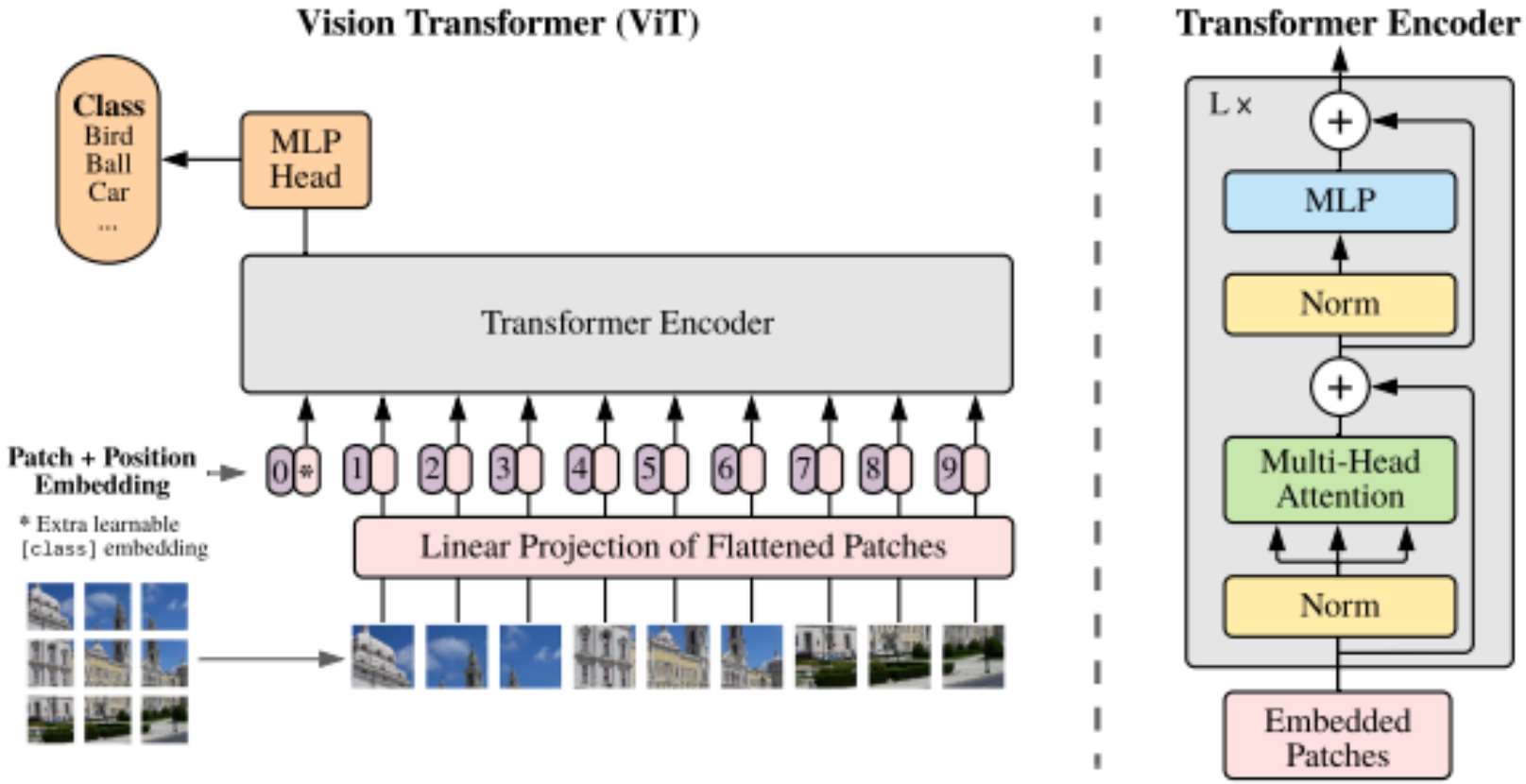


Feature Extraction (Convolution, Pooling, Activation)

Classification (Fully Connection, Activation)

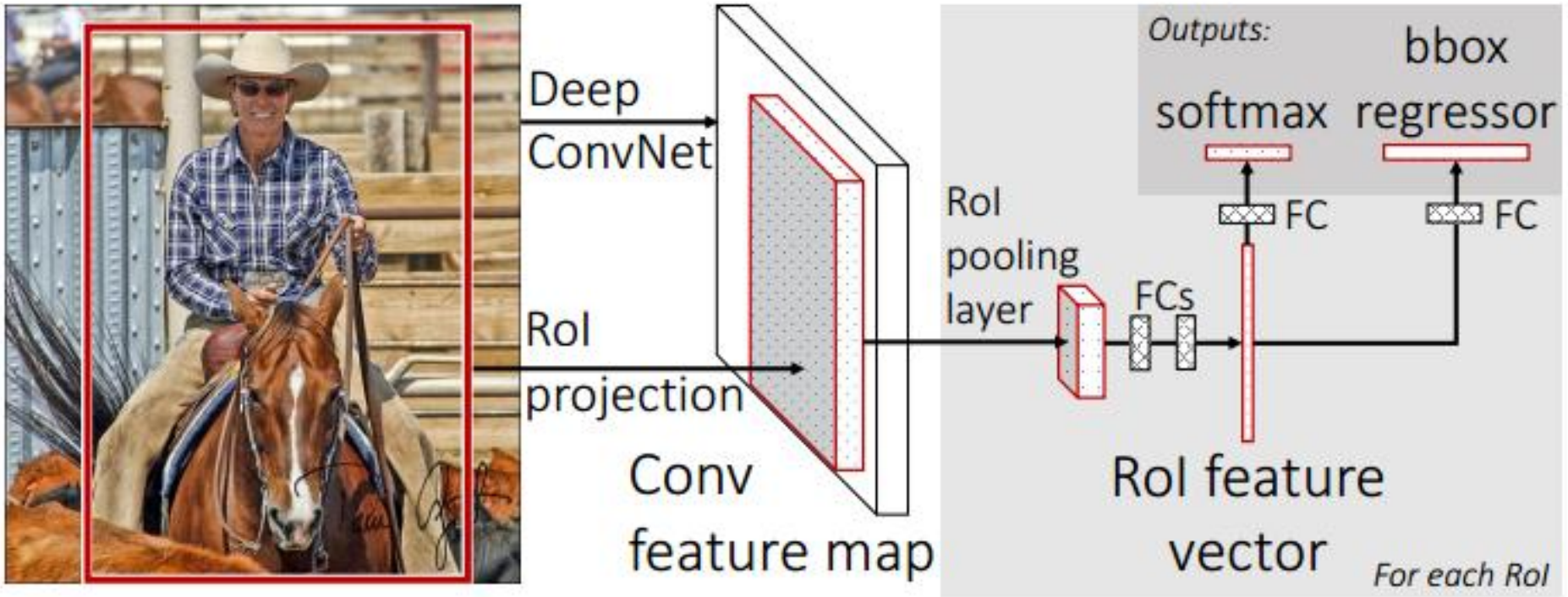
Recognition

- ViT



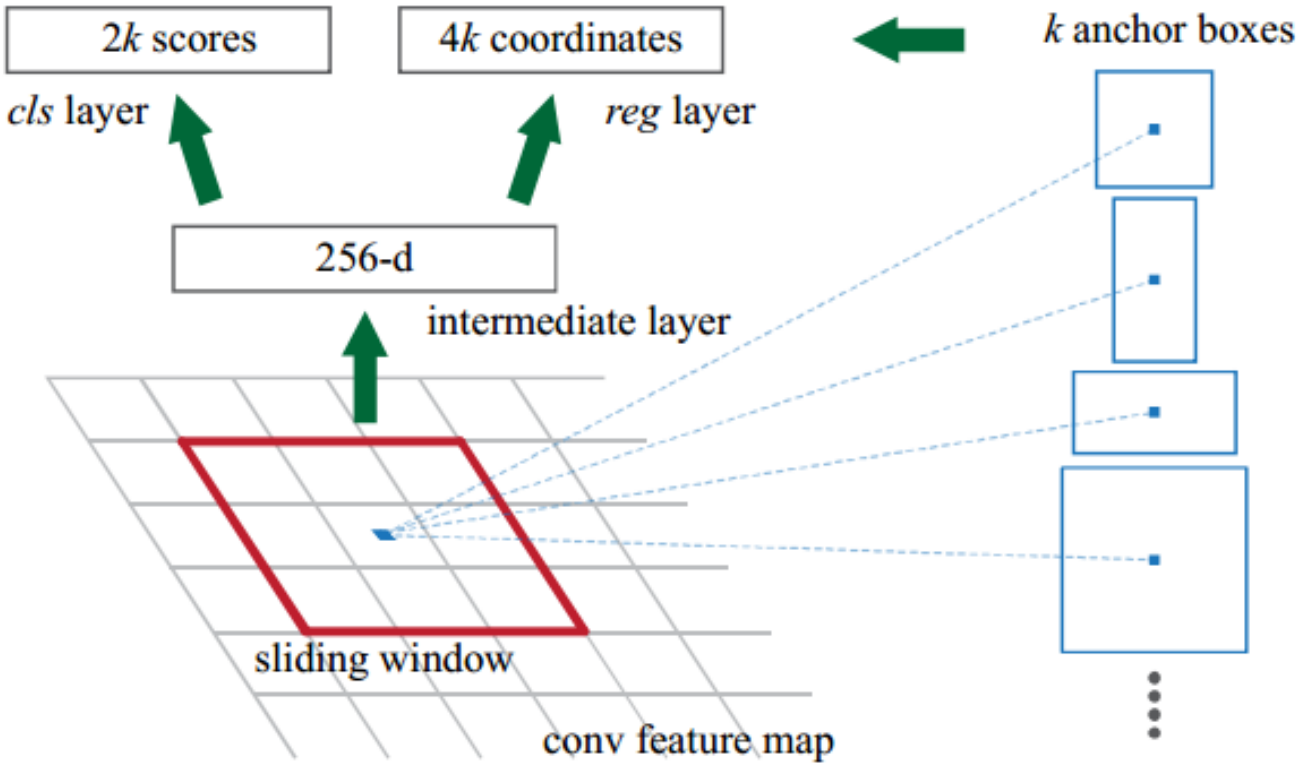
Detection

- Faster R-CNN



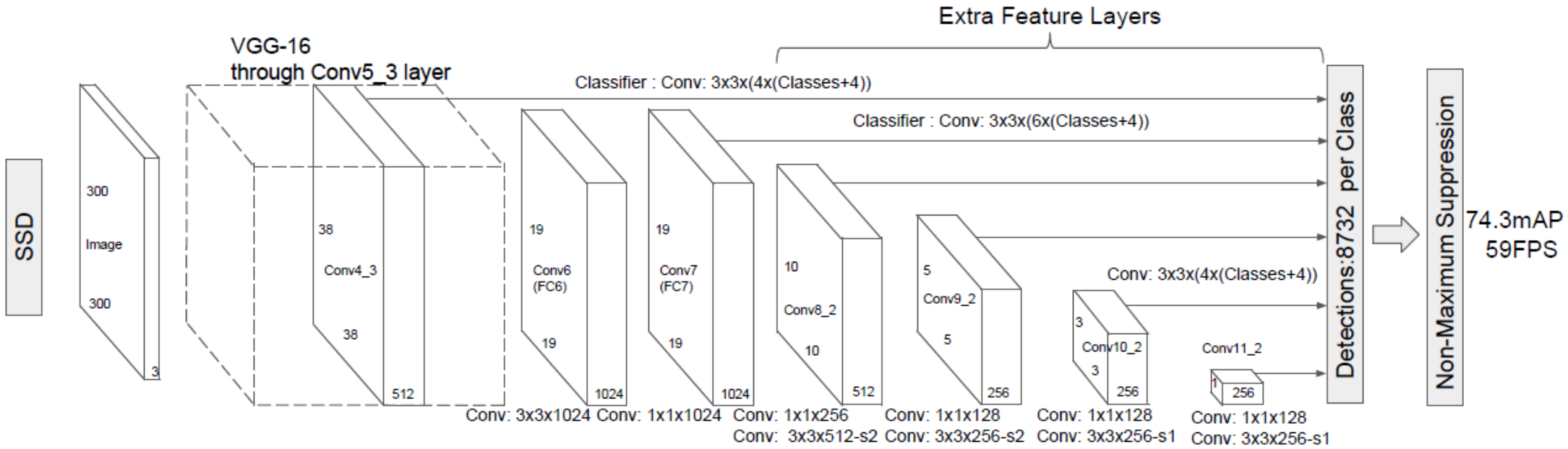
Detection

- Faster R-CNN



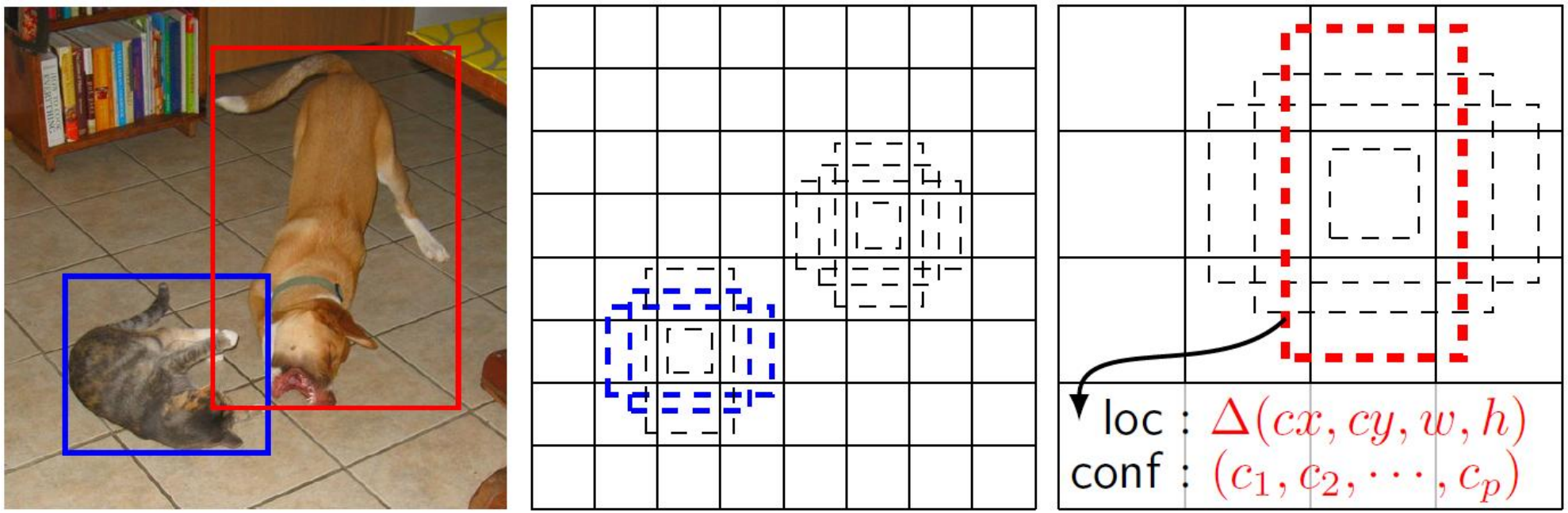
Detection

- SSD：Single Shot MultiBox Detector



Detection

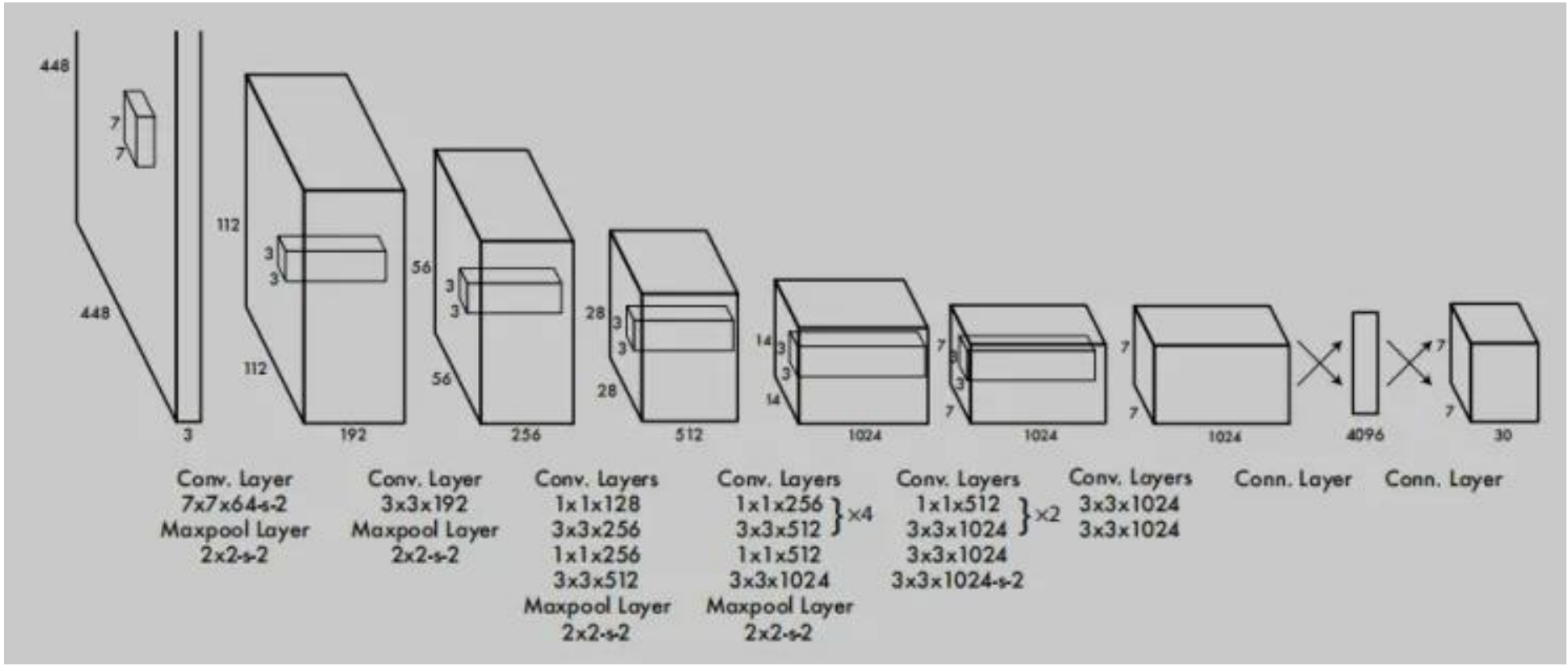
- SSD：Single Shot MultiBox Detector



(a) Image with GT boxes (b) 8×8 feature map (c) 4×4 feature map

Detection

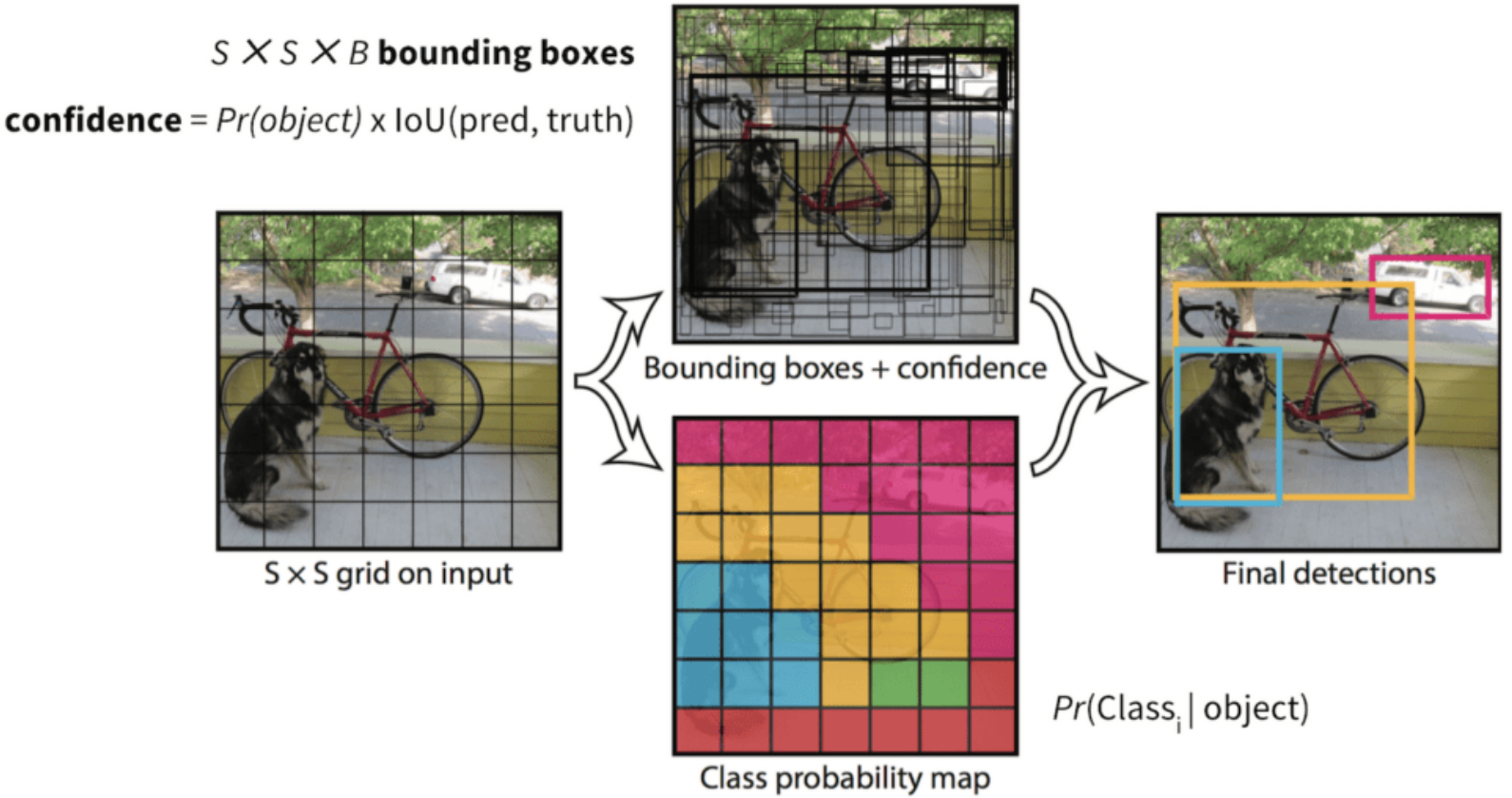
- YOLO : You Only Look Once



YOLO V1

Detection

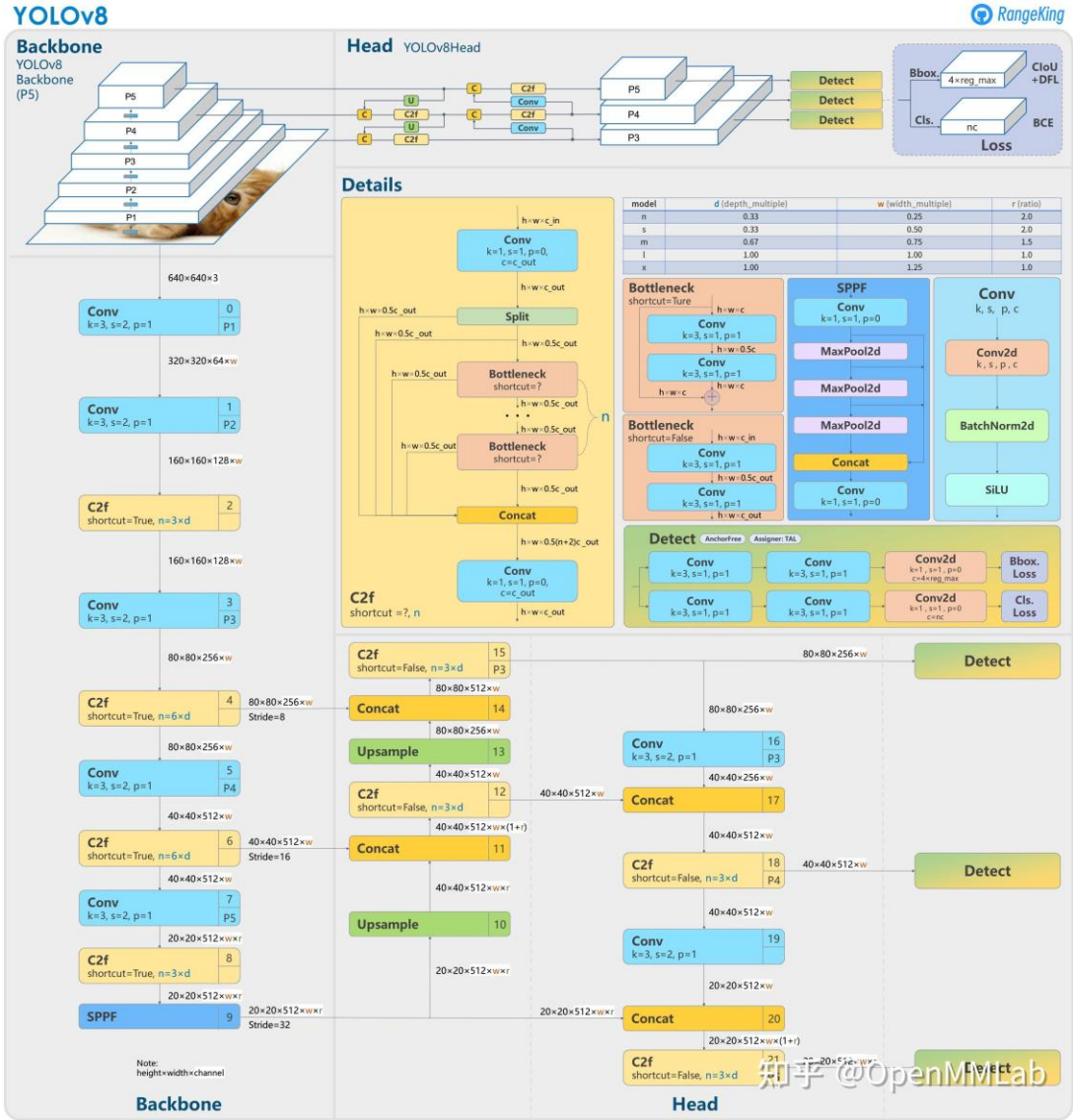
- YOLO : You Only Look Once



YOLO V1

Detection

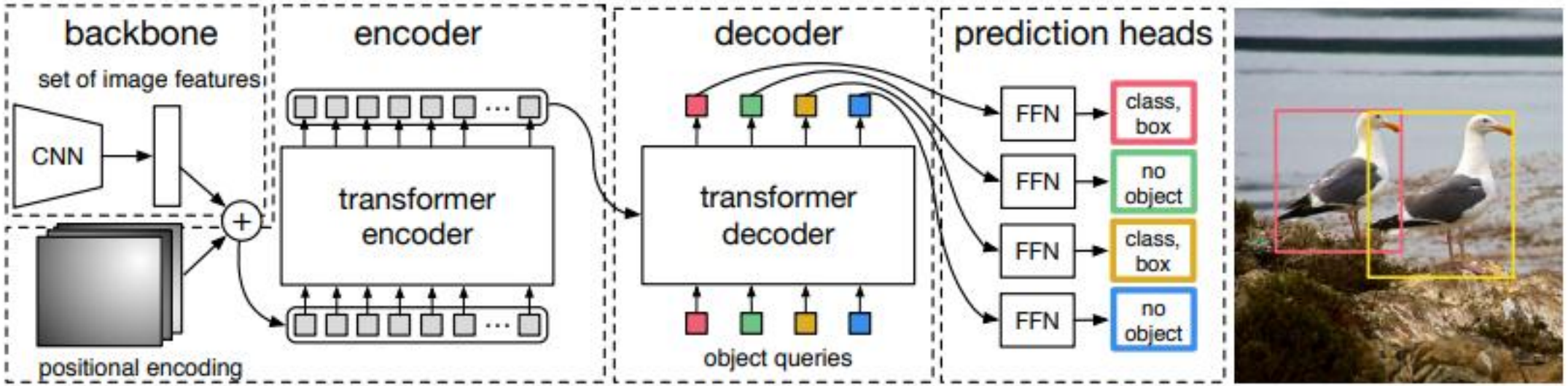
- YOLO：You Only Look Once



YOLO V8

Detection

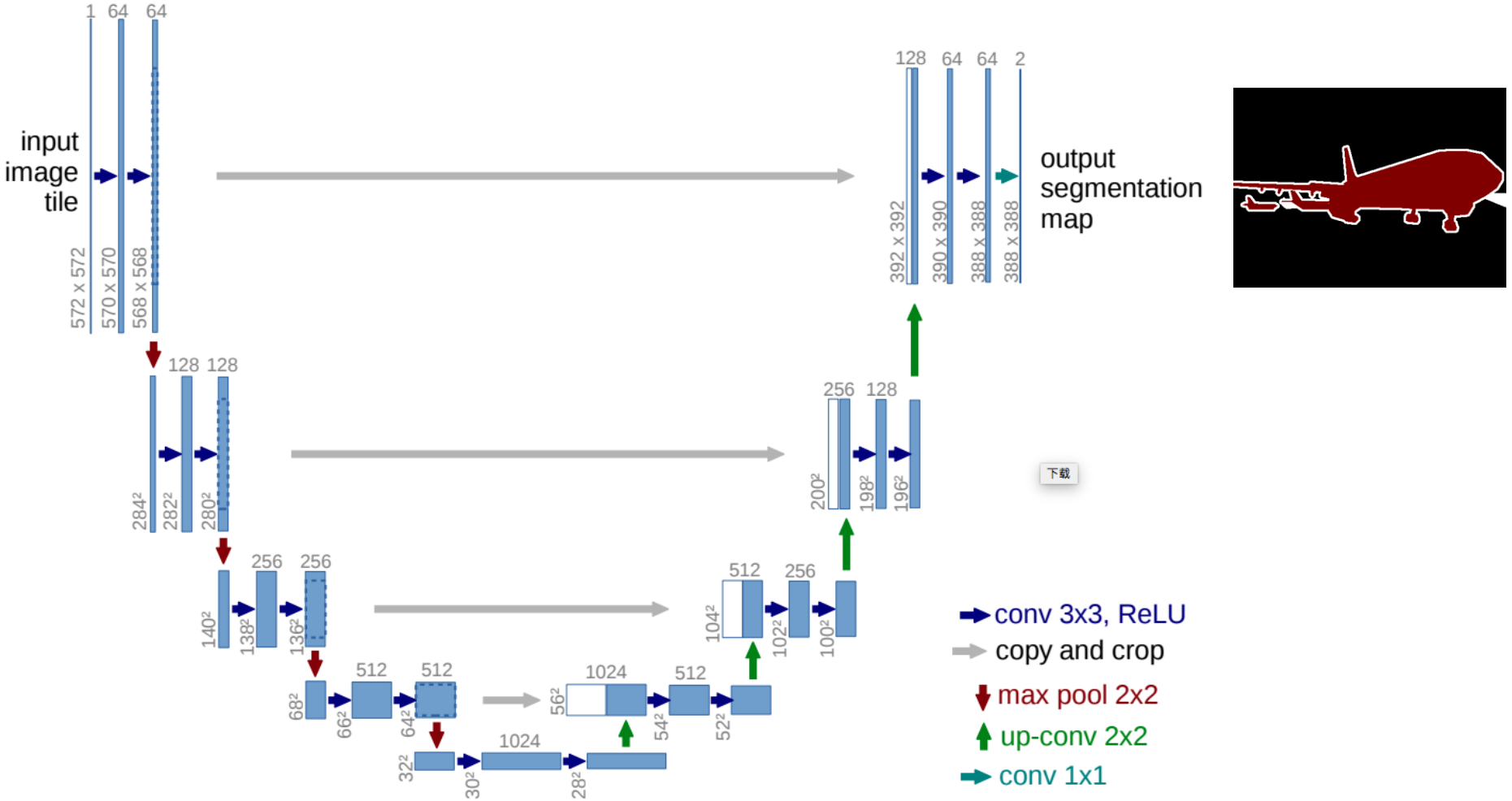
- DETR



Segmentation

- U-Net





Deep Learning for Semantic Segmentation

Outline

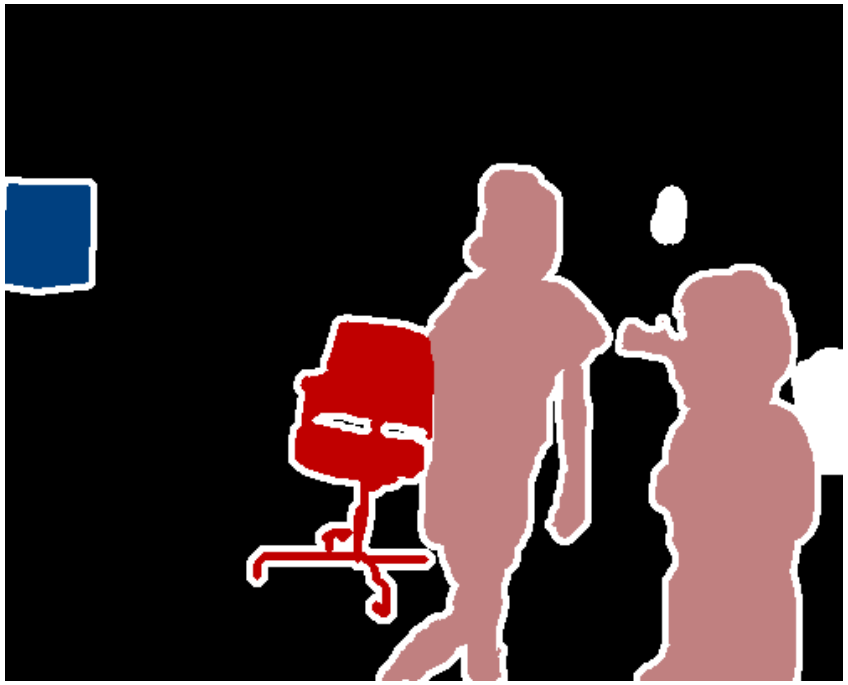
- Semantic Segmentation Review
- Discriminative Feature Network
- Bilateral Segmentation Network

Semantic Segmentation Review

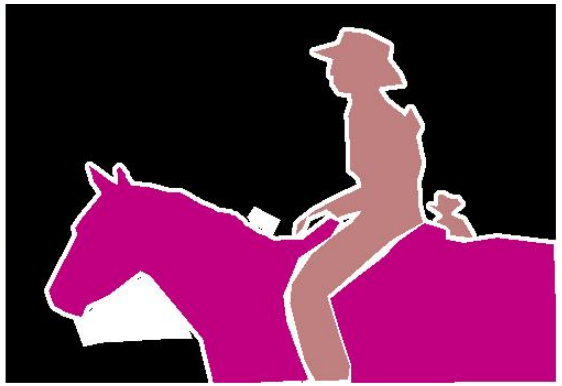
What is Semantic Segmentation ?

- Pixel-level
- Do not differentiate instances

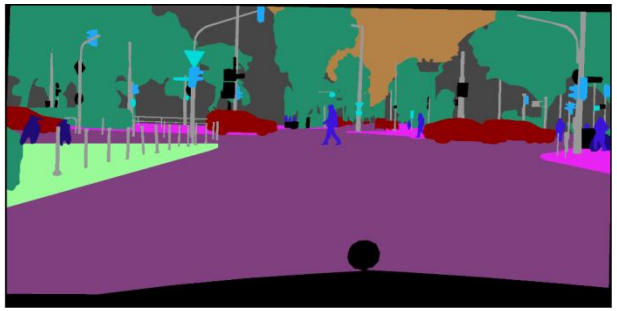




Datasets Overview

PASCAL VOC

Cityscapes




ADE 20K





Mapillary

Discriminative Feature Network

Challenges

- Intra-class Inconsistency
  - same semantic label
  - different appearances
- Inter-class Indistinction
  - different sematic label
  - similar appearances



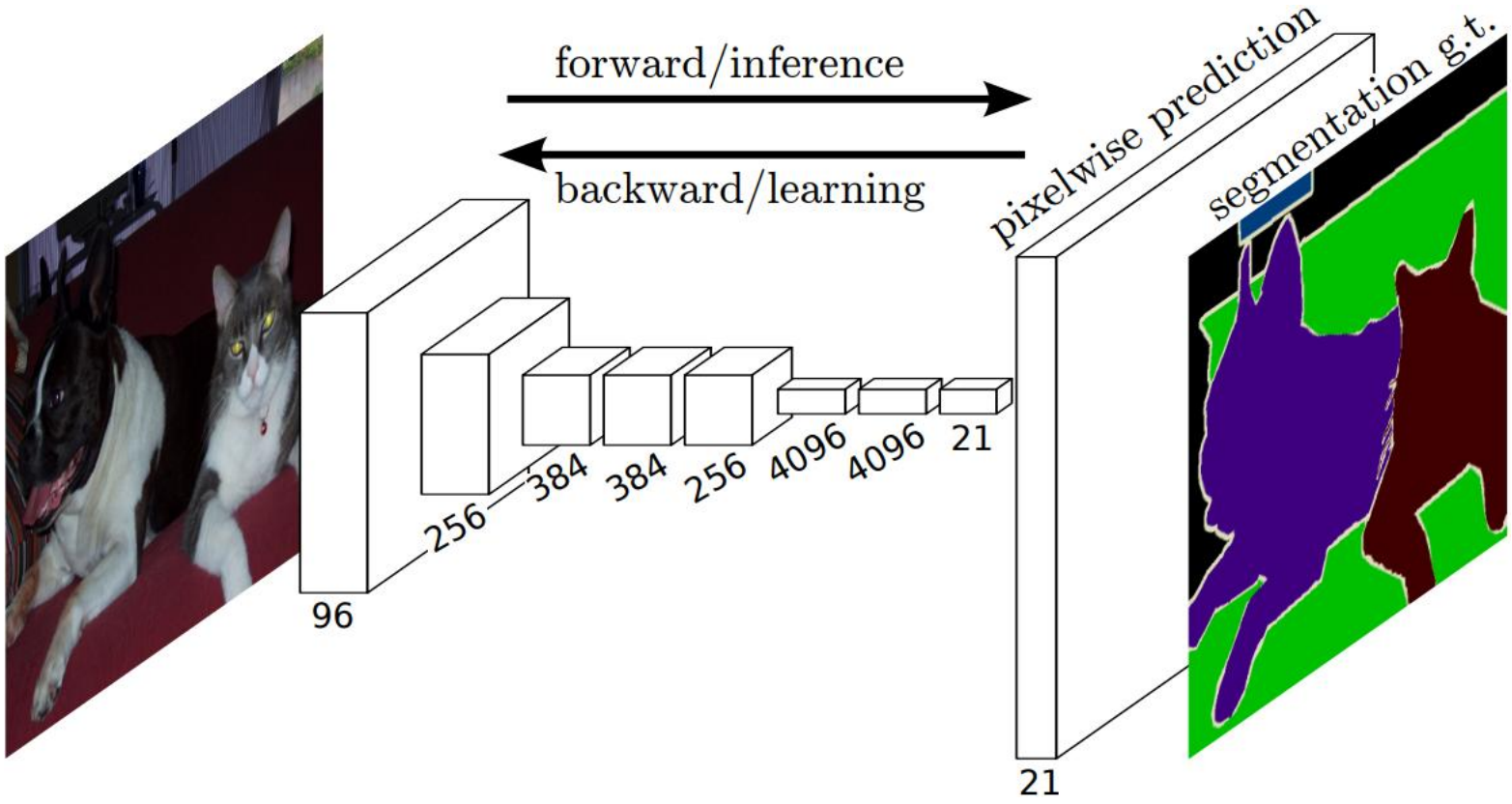
Intra-class Inconsistency



Inter-class Indistinction

Traditional view

- Assign a semantic label to each single pixel

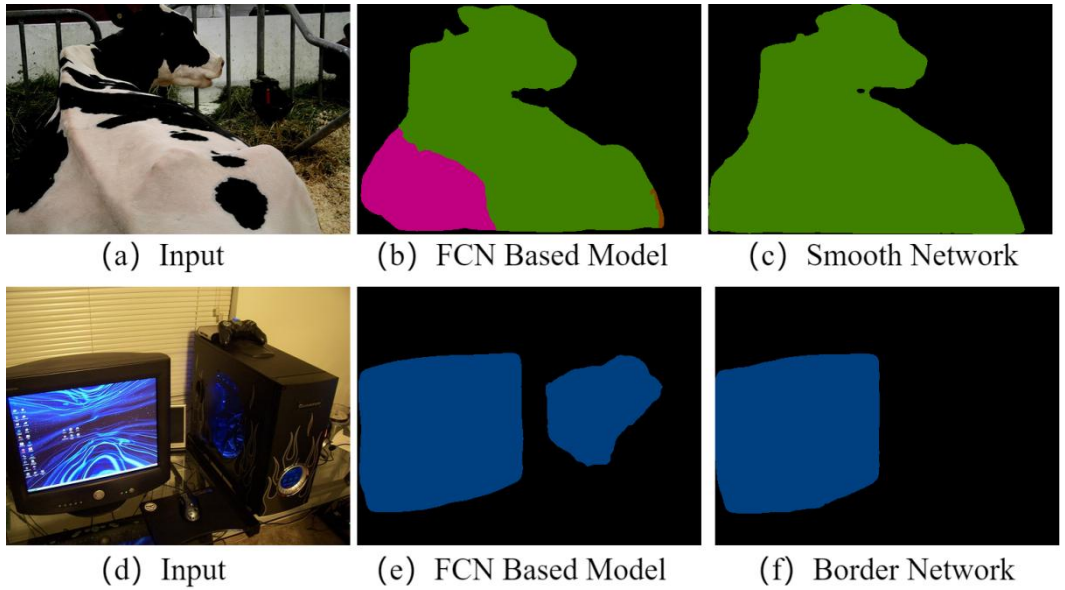


Rethink

- Macroscopic view: assign a consistent semantic label to a category of things
  - Regarding each category of pixels as a whole
    - Intra-class consistency
    - Inter-class variation

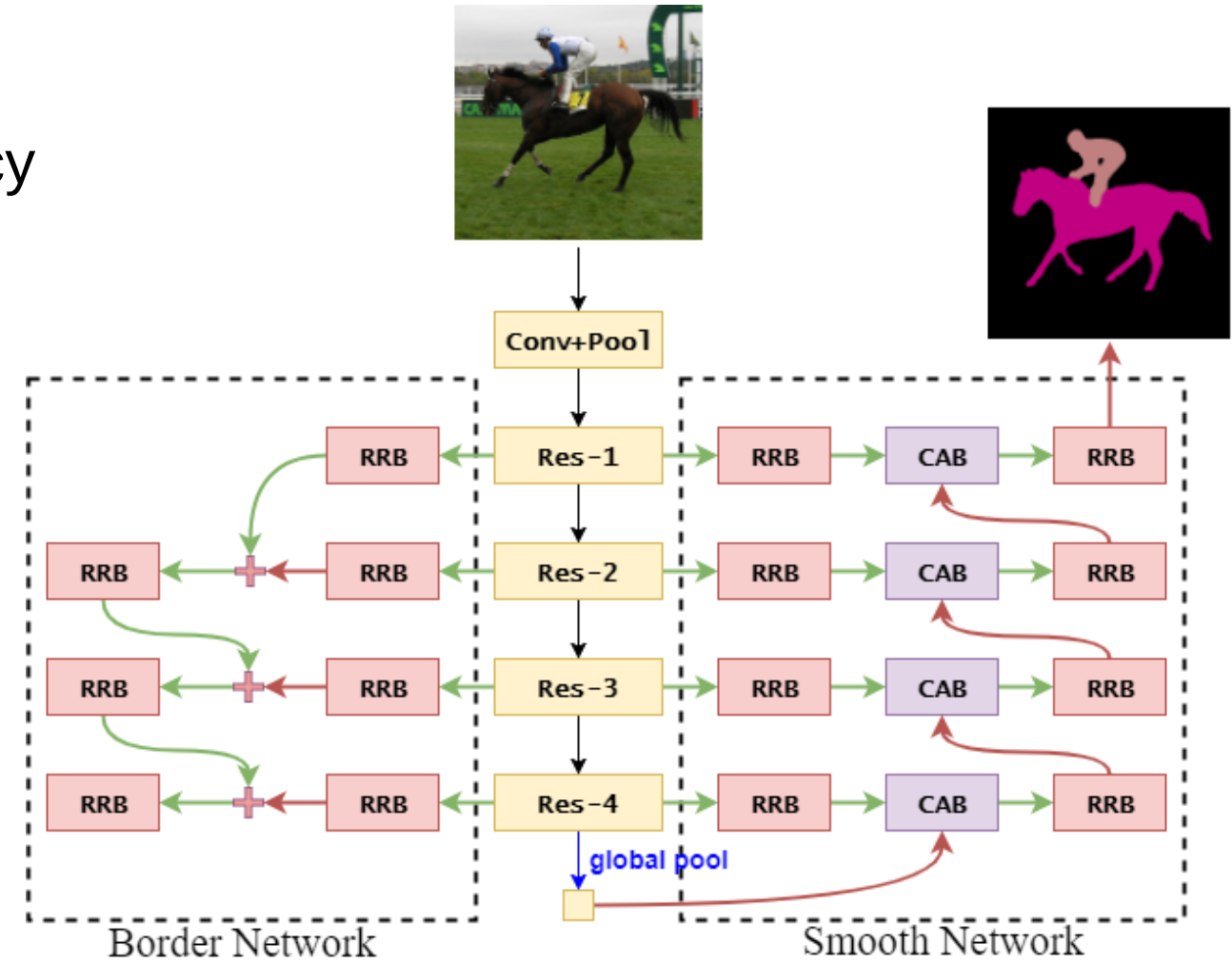


Discriminative Feature



Discriminative Feature Network

- Smooth Network
  - Address: intra-class inconsistency
- Border Network
  - Address: inter-class indistinction

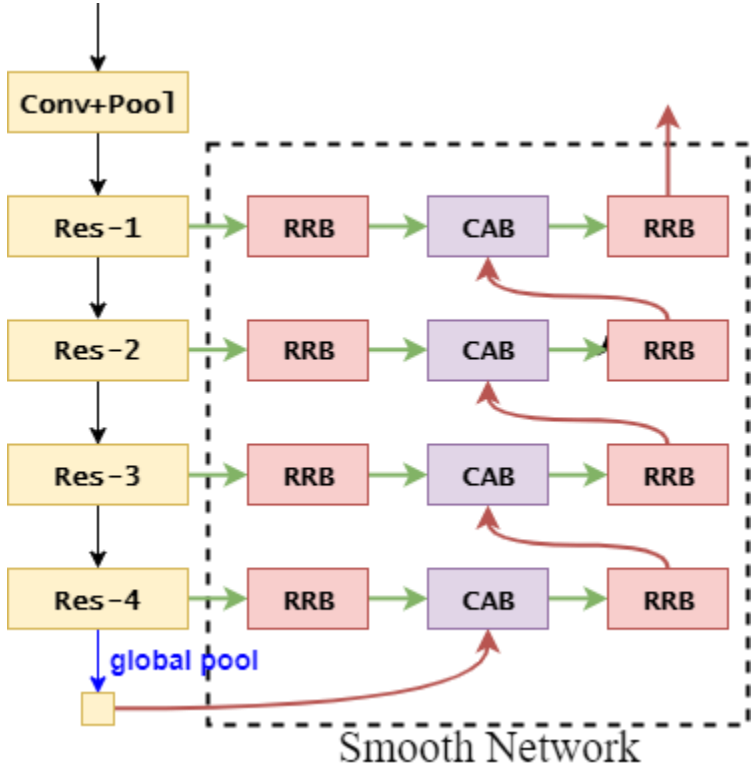


Smooth Network

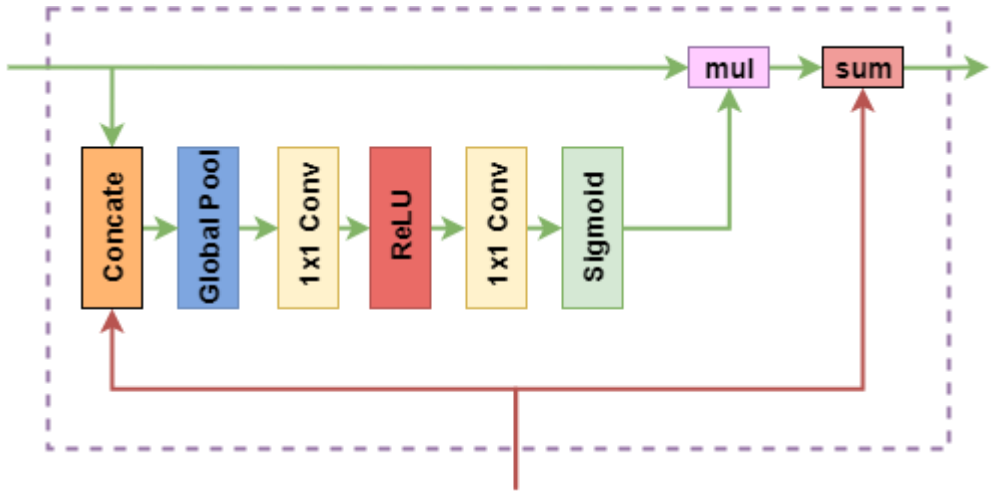
- Intra-Class Inconsistency:
  - Multi-scale and global context
    - Enable the patches to know the image from local to global
  - Selection is important
    - Different scales of receptive fields produce the features with different extents of discrimination
    - Selecting the discriminative and effective features

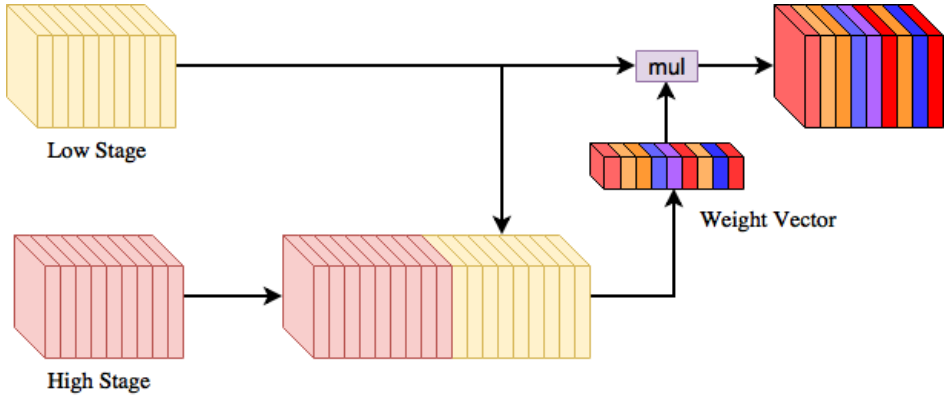
Smooth Network

- Structure
  - U-shape + global context → V-shape
    - Introduce global context + multi-scale context
  - Channel Attention Block (CAB)
    - Change the feature weights to enhance consistency
  - Refinement Residual Block (RRB)



Channel attention block





$$y = F(x; w) = \sum_{i=1,j=1}^{D} \omega_{i,j} x_{i,j}$$

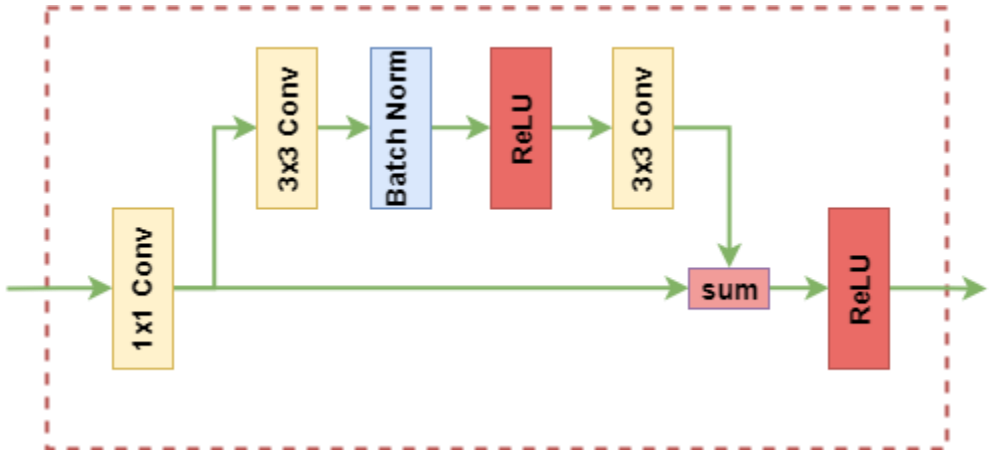
$$y = (y_1, y_2, \cdots, y_n)$$

$$\sigma_i(y_i) = \frac{\exp(y_i)}{\sum_{j=1}^{n} \exp(y_j)}$$

$$\alpha = Sigmoid(x; w)$$

$$\bar{y} = \alpha y = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

Refinement Residual Block

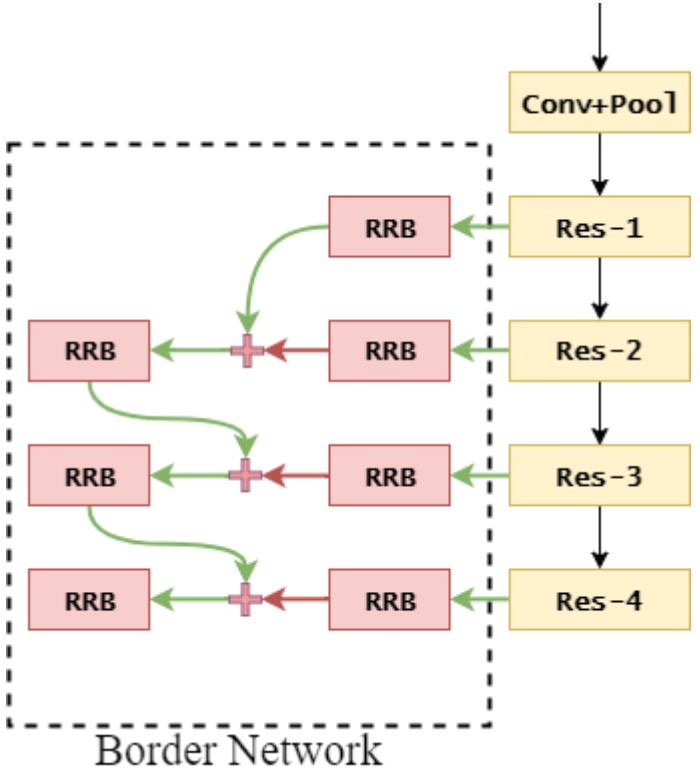


Border Network

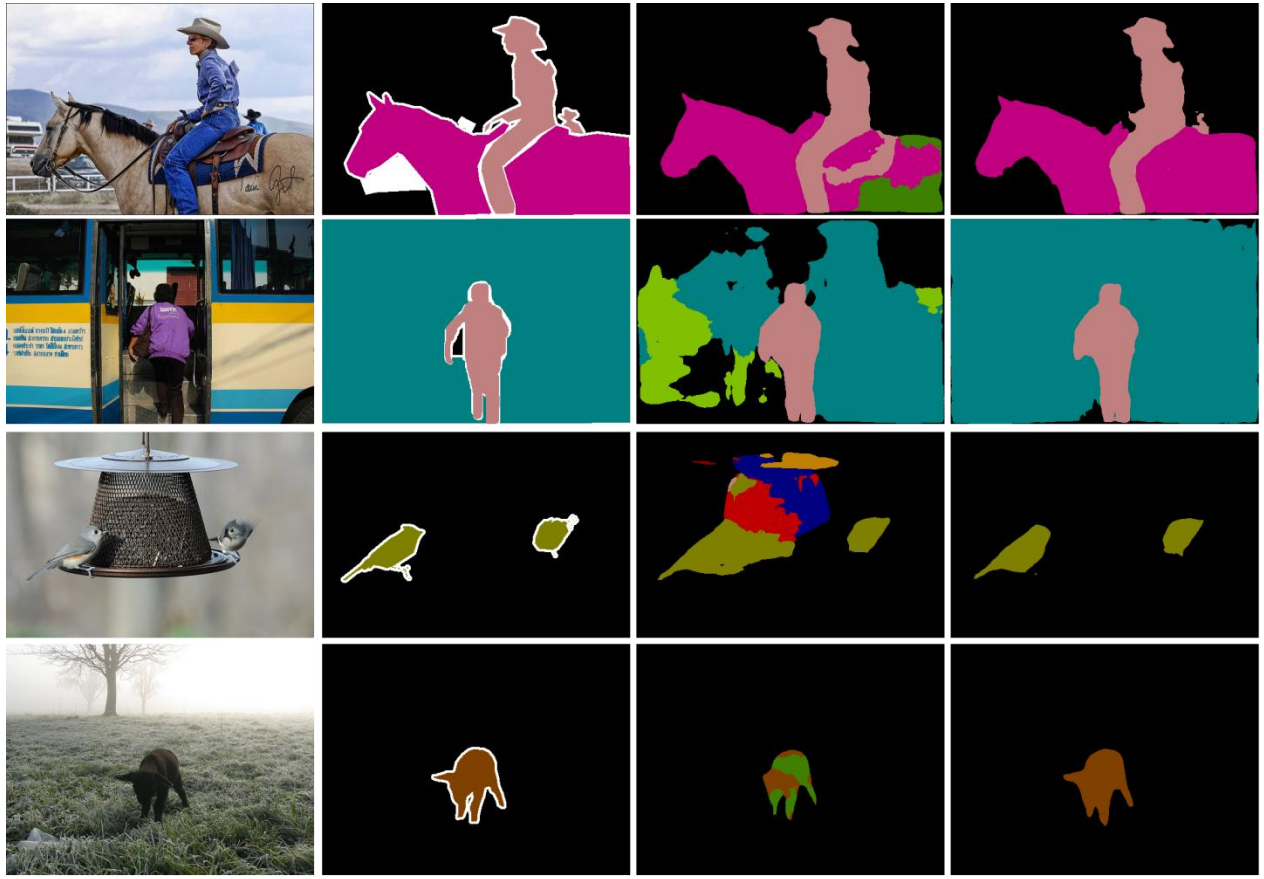
- Inter-class Indistinction
  - Semantic Boundary to guide feature learning
  - Low stage: detailed edge information
    High stage: semantic boundary information

Border Network

- Structure
  - Bottom-up Network
  - Stage-wise Refinement
  - Deep Supervision + Focal-Loss
    - imbalance of the positive and negative samples
  - Refinement Residual Block

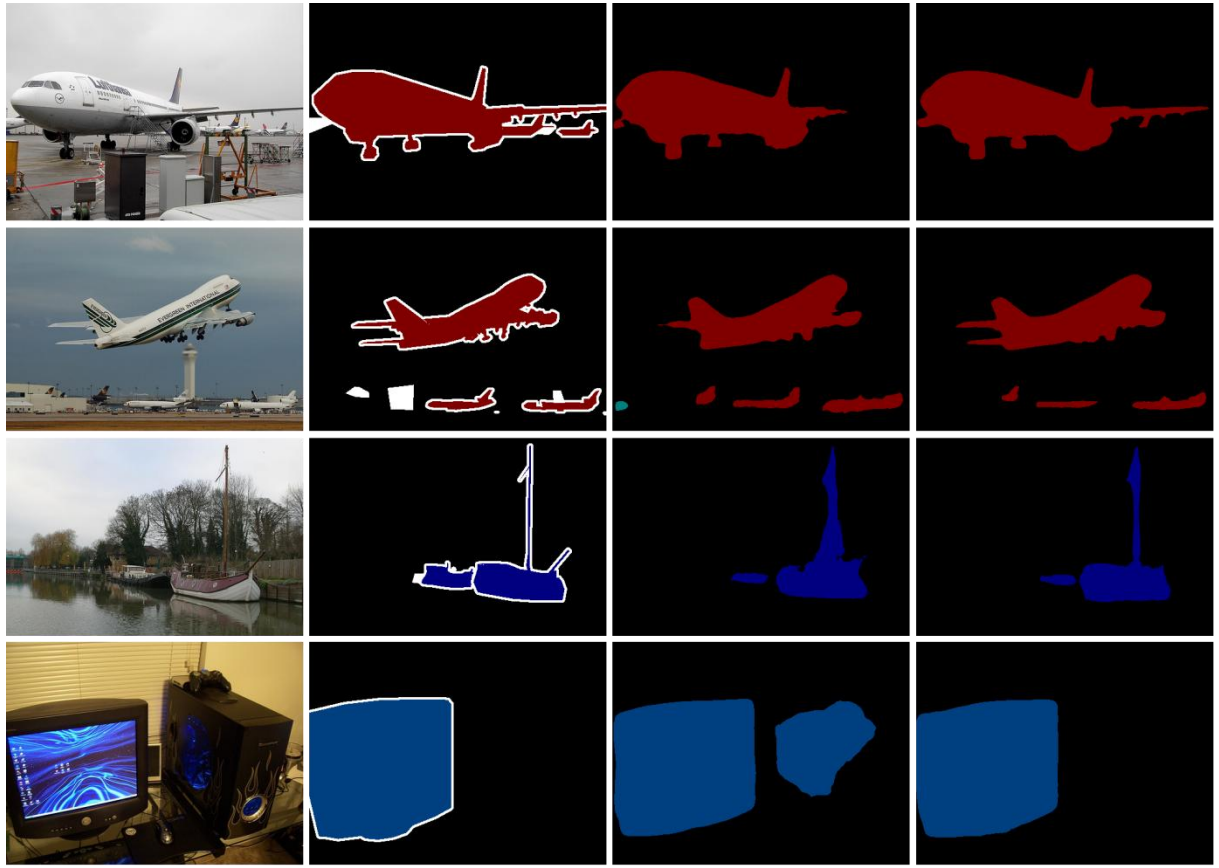


Ablation Study



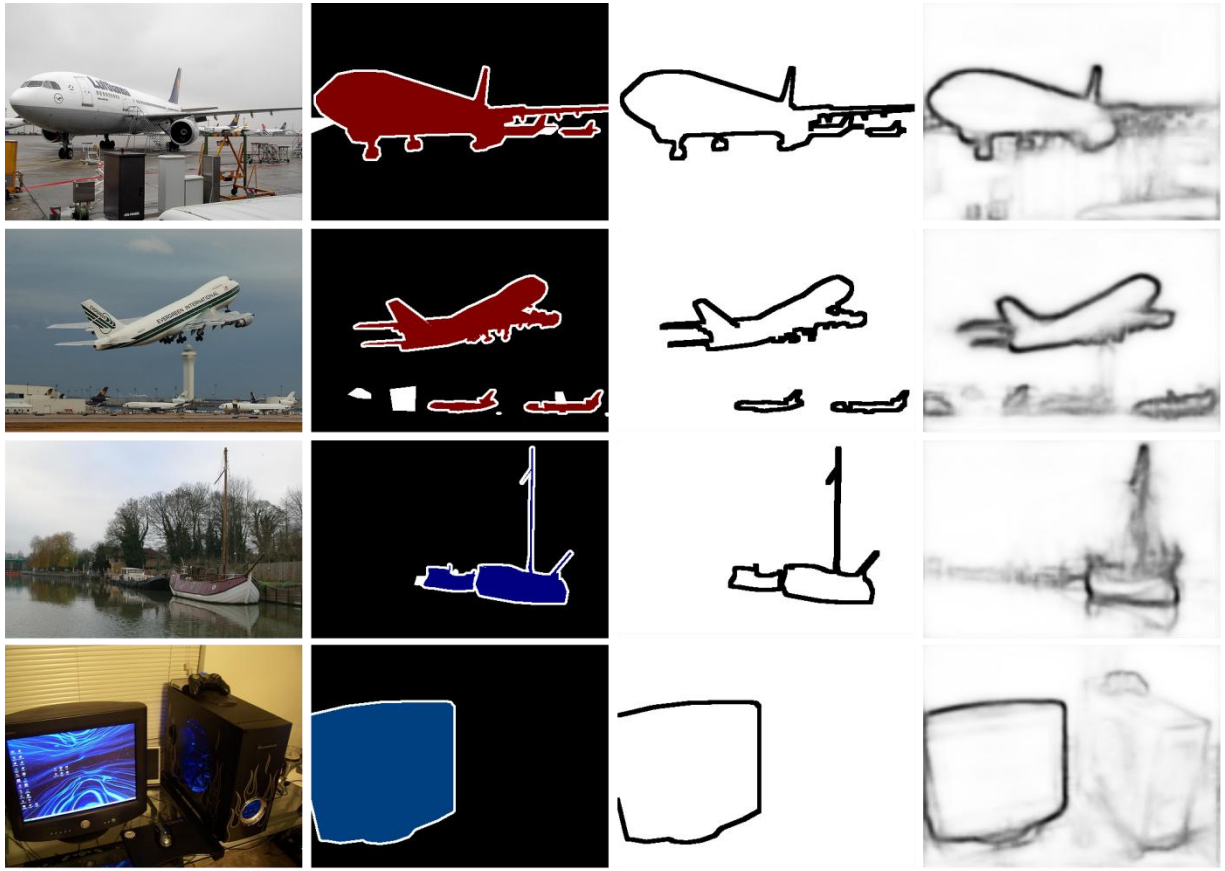
(a) Input (b) GroundTruth (c) Before SN (d) After SN

Ablation Study



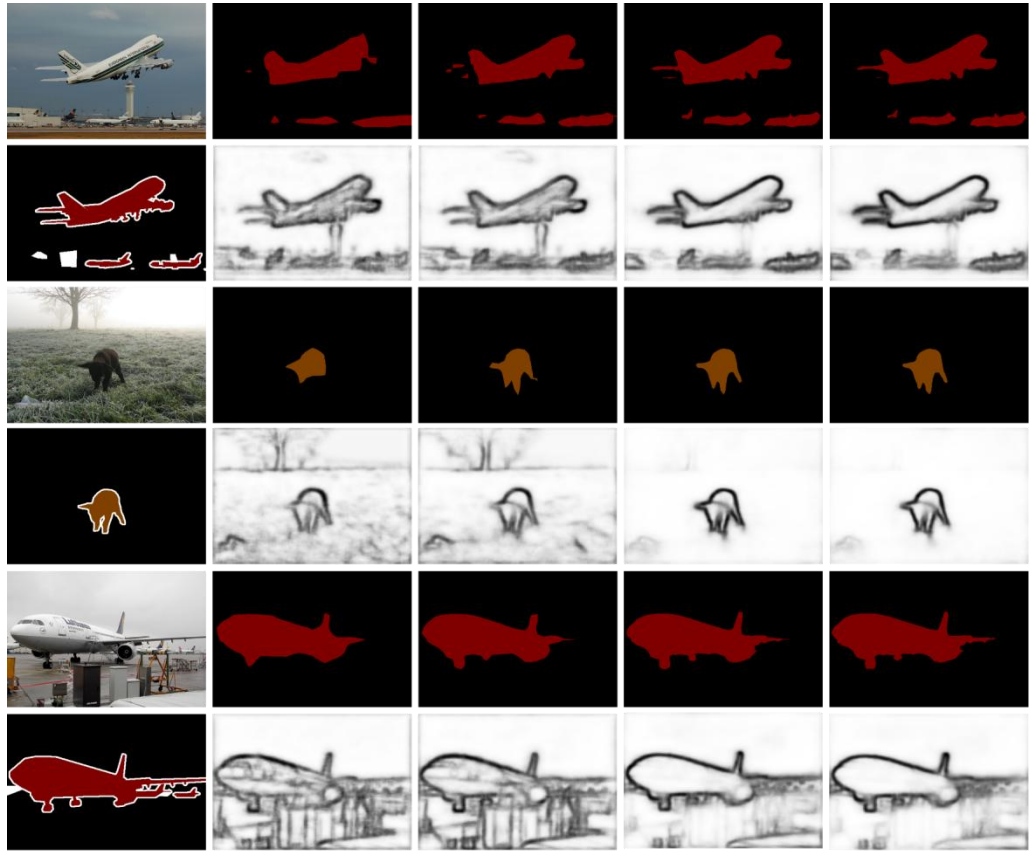
(a) Input (b) GroundTruth (c) Before BN (d) After BN

Boundary Output of Border Network

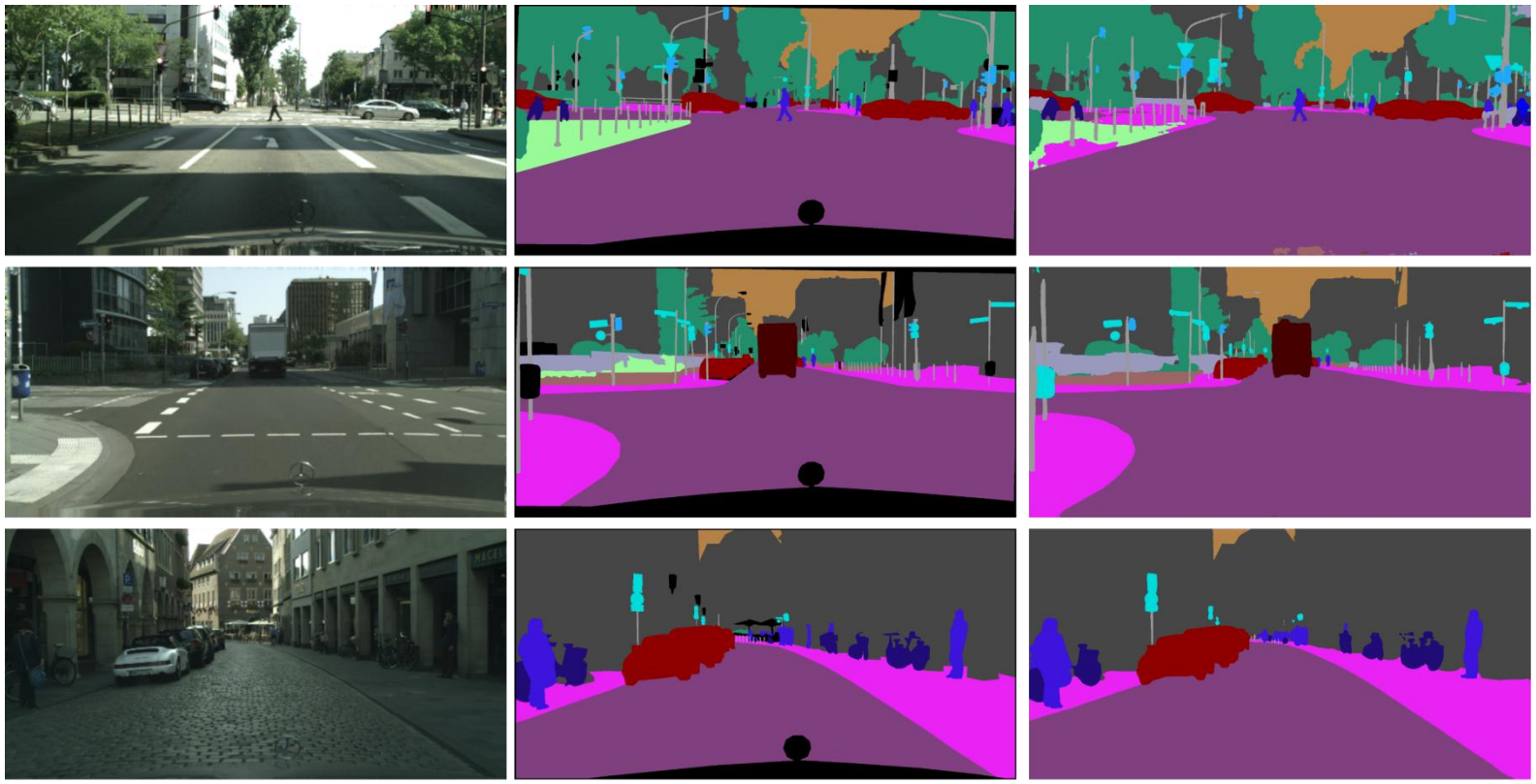


(a) Input (b) GroundTruth (c) Boundary (d) Prediction

Stage-wise Refinement



Example Results on Cityscapes



(a) Input (b) GroundTruth (c) DFN

Results on PASCAL VOC

Method	Mean IOU(%) w/o COCO
FCN	62.2
Zoom-out	69.6
ParseNet	69.8
Deeplab v2-CRF	71.6
DPN	74.1
Piecewise	75.3
LRR-CRF	75.9
PSPNet	82.6
Ours	**82.7**

Method	Mean IOU(%) w/ COCO
Adelaide_VeryDeep_FCN	79.1
LRR_4x_ResNet-CRF	79.3
Deeplab v2-CRF	79.7
Deep Layer Cascade	82.7
DUC	83.1
GCN	83.6
RefineNet	84.2
PSPNet	85.4
Deeplab v3	85.7
Ours	**86.2**

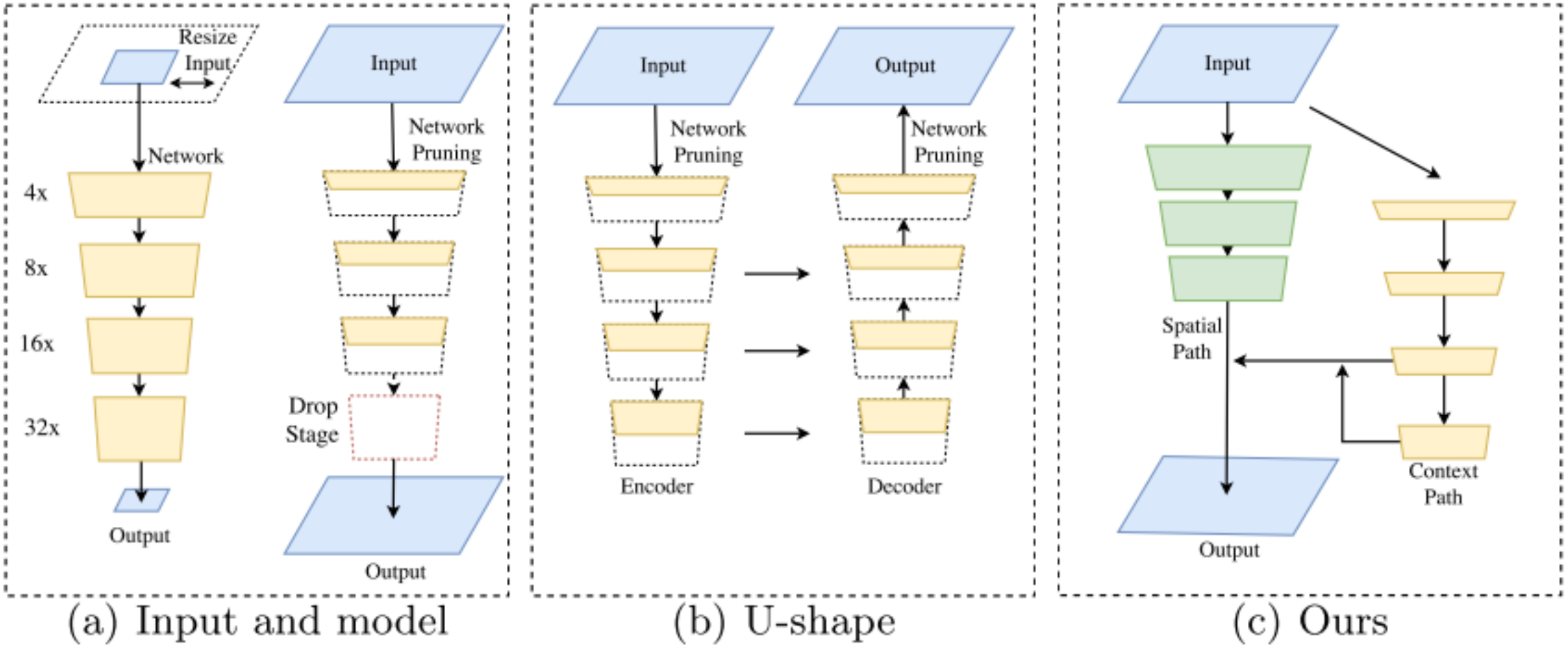
Results on Cityscapes

Method	Mean IOU(%) w/o coarse
FCN	65.3
DPN	66.8
LRR	69.7
Deeplab v2-CRF	70.4
Piecewise	71.6
RefineNet	73.6
GlobalLocalRefinement	77.3
DUC	77.6
SegModel	78.5
Ours	**79.6**

Method	Mean IOU(%) w/ coarse
Deeplab v2-CRF	70.4
Adelaide_context	71.6
LRR-4x	71.8
GCN	76.9
SegModel	79.2
DUC	80.1
PSPNet	80.2
Ours	**80.3**

Bilateral Segmentation Network

Previous Methods



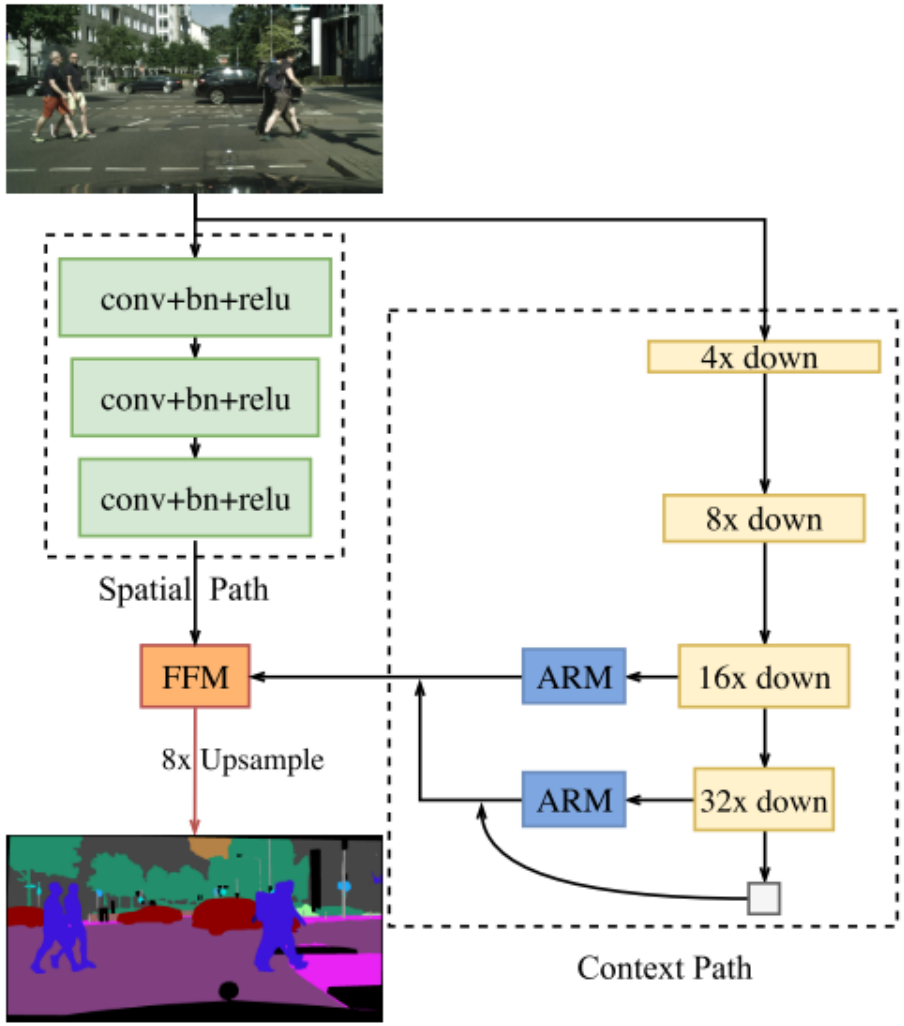
Previous Methods

- Problem:
  - Loss of spatial details
  - Sacrifice the spatial capacity

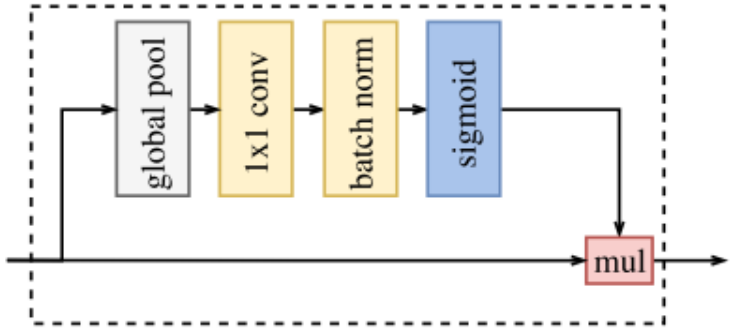
} Compromise the accuracy to speed

- Solution:
  - Spatial Path: spatial information preservation
  - Context Path: receptive field offering

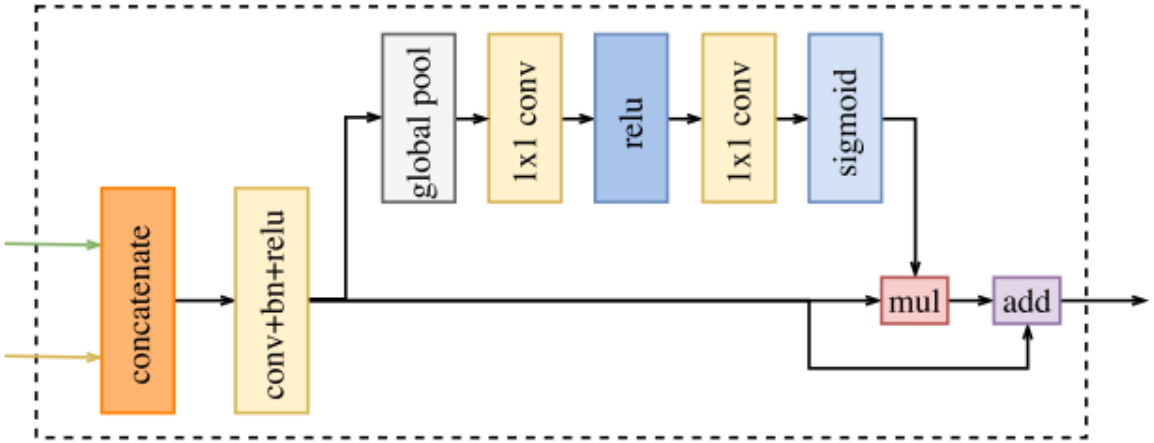
Bilateral Segmentation Network



(a) Network Architecture



(b) Attention Refinment Module



(c) Feature Fusion Module

Ablation Study on Cityscapes

Method	Mean IoU(%)
CP	66.01
CP+SP(Sum)	66.82
CP+SP(FFM)	67.42
CP+SP(FFM)+GP	68.42
CP+SP(FFM)+ARM	68.72
CP+SP(FFM)+ARM+GP	71.40

CP: Context Path
SP: Spatial Path
FFM: Feature Fusion Module
ARM: Attention Refinement Module
GP: Global Average Pooling

Speed Analysis on Cityscapes

Method	BaseModel	GFLOPS	Parameters
SegNet	VGG16	286.0	29.5M
Enet	From scratch	3.8	0.4M
Ours	Xception39	2.9	5.8M
Ours	Res18	10.8	49.0M

FLOPS & Parameters

Method	NVIDIA Titan X						NVIDIA Titan XP					
	640x360		1280x720		1920x1080		640x360		1280x720		1920x1080	
	ms	*fps*	*ms*	*fps*	*ms*	*fps*	*ms*	*fps*	*ms*	*fps*	*ms*	*fps*
SegNet	69	14.6	289	3.5	637	1.6	-	-	-	-	-	-
Enet	7	135.4	21	46.8	46	21.6	-	-	-	-	-	-
Ours	**5**	**203.5**	**12**	**82.3**	**24**	**41.4**	**4**	**285.2**	**8**	**124.1**	**18**	**57.3**
Ours	8	129.4	21	47.9	43	23	5	205.7	13	78.8	29	34.4

Speed Comparison

Speed Analysis on Cityscapes

Method	BaseModel	Mean IOU(%)		FPS
		val	*test*	
SegNet	VGG16	-	56.1	-
Enet	From scratch	-	58.3	-
SQ	SqueezeNet	-	59.8	-
ICNet	PSPNet50	67.7	69.5	30.3
DLC	Inception-ResNet-v2	-	71.1	-
Two-column Net	Res50	74.6	72.9	14.7
Ours	Xception39	69.0	68.4	**105.8**
Ours	Res18	**74.8**	**74.7**	65.5

Speed and Accuracy Comparison

Accuracy Analysis on Cityscapes

Method	BaseModel	Mean IOU(%)	
		val	*test*
DeepLab	VGG16	-	63.1
FCN-8s	VGG16	-	65.3
Adelaide	VGG16	-	66.4
Dilation10	VGG16	68.7	66.4
LRR	VGG16	70.0	69.7
DeepLab-v2+CRF	Res101	71.4	70.4
RefineNet	Res101	-	73.6
DUC	Res152	76.7	76.1
PSPNet	Res101	-	78.4
Ours	Xception39	72.0	71.4
Ours	Res18	78.6	77.7
Ours	Res101	**80.3**	**78.9**

Accuracy Comparison

Accuracy Analysis on COCO Stuff

Method	BaseModel	Mean IOU(%)	Pixel Accuracy(%)
DeepLab-v2	VGG-16	24.0	58.2
Ours	Xception39	22.8	59.0
Ours	Res18	28.1	63.2
Ours	Res101	**31.3**	**65.5**

Accuracy Comparison on COCO Stuff

The end !